Testing a predictive biomarker under drug carryover in aggregated N-of-1 trials: a staged simulation study

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Abstract

Background. Aggregated N-of-1 trials pool repeated within-patient on/off contrasts to validate predictive biomarkers. Drug carryover attenuates the biomarker-by-treatment interaction. The analyst must decide how to represent residual exposure, how to size the test, and whether the half-life must be known.

Methods. In an exploratory simulation under the structure of a companion paper's data-generating process, with a predictive-only biomarker, we compared an analysis ladder in six scenario families. Rung 0 is the starting analysis (A0); rung 1 a binary drug indicator with `corCAR1` residual correlation (A1); rung 2 adds CR2 cluster-robust inference (A2); rung 3 an exposure-weighted predictor at the true half-life (A3, an oracle); rung 4 chooses the half-life by AIC on the model without the interaction (A4).

Results. Pooled over three designs in the base grid, A0 rejected the null at 0.110 and A1 at 0.038. CR2 moved size to 0.055. In the Hybrid design A4 exceeded A2, the CR2 binary indicator, by 1.3 points on average over 40 alternative cells and by up to 7.1 at long half-lives. It was below A2 in 6 cells, by at most 1.0 point. In the run with independent seeds, the size of A4 minus that of the oracle was -0.0008 (90 percent interval -0.0033 to +0.0018), equivalent within ±0.01. For 80 percent power at a 1-week half-life, a Hybrid trial analyzed with A4 needs about 73 patients at $c_{bm} = 0.4$ and 292 at 0.2.

Conclusions. For a Hybrid trial with a biomarker known to be predictive only, under this process at $N \leq 70$, A4 is correctly sized and needs no half-life. Its gain over the binary indicator is modest except at long half-lives. Even the oracle recovers only part of the power that carryover removes, so the rest has to be met by the design. Whether a biomarker is also prognostic is rarely known before a trial. In that case the analysis of the companion paper applies; this paper's results bear on its choice of half-life and its CR2 test. The results are conditional on the correlation encoding and one parameter set.

Keywords: N-of-1 trials; carryover; predictive biomarker; cluster-robust inference; post-selection inference; simulation study.

1 Introduction

In a single N-of-1 trial, one patient is crossed repeatedly between active treatment and a comparator. The within-patient contrast between on-drug and off-drug periods estimates that patient's individual treatment effect. An aggregated N-of-1 trial pools many such series. Pooling allows population-level inference at sample sizes smaller than a parallel-group trial would require (1). The estimand here is the biomarker-treatment interaction, that is, whether a baseline covariate B_i modifies the treatment effect. We evaluate analysis approaches for this estimand in the Hybrid N-of-1 design defined in Section 2.2.

The main inferential complication in N-of-1 trials is carryover. A drug's pharmacological effect does not stop at discontinuation. It decays as a function of the drug's half-life. Observations taken after discontinuation therefore still carry part of the treatment effect. An exposure regressor that ignores this decay is a mismeasured version of what the patient received. The consequences are twofold. First, the interaction contrast is attenuated. Occasions that still carry a large fraction of the drug effect are coded the same as occasions that carry none. This biases the biomarker-by-exposure coefficient toward zero, as any covariate measured with error would. Second, those occasions vary in a way the model does not represent. The unexplained variation inflates the standard error of the same coefficient. Power falls for both reasons.

This is a loss of power, not of validity. Under the null, $c_{bm} = 0$, there is no biomarker-response association at any level of residual exposure. A miscoded regressor then has no interaction to distort, and test size does not depend on the exposure coding. The conservatism reported in Section 3.2.2 has a separate origin in the working covariance and applies equally to every specification. Carryover misspecification is therefore a question of efficiency. Rung 3 of the study is therefore framed in terms of recovered power.

A companion study (2) builds on the data-generating process of Hendrickson et al. (1). In that process the interaction is encoded as a correlation between the biomarker and the drug response. The correlation is present on drug and decays off drug: $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$. We call this the correlation encoding of the interaction. The companion study shows that carryover reduces power under this encoding, and that the reduction depends on the design. The loss under a given analysis is governed by the on/off correlation difference, that is, the on-drug correlation minus the mean off-drug correlation, which the analysis's exposure coding determines. At a half-life of one week in this study's factorial grid, the binary indicator loses a fifth to two thirds of its power under the Hybrid and OL+BDC designs. It loses 7 to 16 percent under the crossover design (Section 3.2.3). We restrict attention to the correlation encoding. The carryover problem is largest there, and the starting DGP uses it. Earlier work under this encoding fixed the decay shape at exponential and evaluated only one analysis specification. Simulation studies of aggregated N-of-1 analyses under carryover exist (3–6). One includes an exponentially decaying wash-in and wash-out term (6). All target the treatment effect rather than the biomarker-by-treatment interaction, which is the estimand here. An interaction test needs about four times the sample of a main-effect test of the same size (7). The correlation encoding is a modeling assumption of this project, not a validated description of any trial. Every recommendation below is conditional on it.

This manuscript shows how a sequence of changes to the analysis model improves both size and power. We call the sequence an analysis ladder. Each rung changes one thing about the analysis. The rungs follow the order in which the questions arise for an analyst. Section 2.4.1 defines the rungs and Table 3 lists them.

Rung 0 is the starting analysis (A0), the analysis used with the starting DGP (1). It fits a random intercept, uses a binary on-drug indicator, and has no residual correlation structure. The first question is whether its test is correctly sized. Rung 1 (A1) keeps the binary indicator and uses an AR(1) residual correlation through `corCAR1` with the working-covariance standard error. A0 and A1 differ in three respects at once. Isolation analyses separate the three and attribute the liberal size of A0 to its missing residual correlation (Section 3.2.1).

The second question concerns the size of the test. The working covariance of the mixed model is a random intercept with an AR(1) residual correlation. It does not match the covariance that the data-generating process induces, and the working-covariance test is conservative. Rung 2 (A2) replaces the working-covariance standard error by a CR2 bias-reduced cluster-robust standard error with Satterthwaite degrees of freedom (8). The rung changes only the standard error, the reference distribution and the p-value.

The third question is how residual exposure should enter the linear predictor. The analyst may ignore the decay and code exposure as on or off, as the starting analysis does (A1 and A2). Or the analyst may represent it by committing to a decay form and half-life in a continuous exposure-weighted predictor. The predictor changes the regressor that carries the interaction. Rung 3 (A3) is the exposure-weighted predictor at the true half-life with CR2 inference. We compare the two positions against exponential and Weibull decay forms in a full factorial.

The fourth question follows from the third position. Rung 3 uses the true half-life, which no real analyst knows, so A3 is an oracle benchmark rather than a candidate analysis. Rung 4 (A4) lets the data choose the half-life by AIC over a candidate grid, from fits of the model without the interaction, and then tests the interaction at the chosen value. The selection step adds a post-selection inference risk (9–11). Leaving the tested term out of the selection limits that risk. We measure what the selection costs against the oracle in size, power and standard-error calibration.

We then stress the ladder in scenario families that a real trial may present. These are a misspecified half-life and decay shape, serial correlation, patient dropout and a patient-specific slope variance.

The biomarker in this study is predictive only: it changes the drug response and nothing else. A biomarker that is also prognostic raises a different question, namely whether the interaction test confuses an association with the natural history or the placebo response with moderation of the drug effect. A prognostic shift that is constant over visits is absorbed exactly by the biomarker main effect of every analysis here. It cannot move the interaction estimate, so it needs no simulation. Prognostic associations that vary over time, and the design-expectancy terms that protect against them, are the subject of a companion paper (12).

Two defects in the simulation code were found during the study and corrected. The first was a pre-exposure coding error in the exposure-weighted predictor that affected the crossover

design. The second was a cross-factor correlation constant that departed from the documented data-generating process. Every result in this manuscript was produced under the corrected code. We follow the ADEMP reporting framework of Morris, White, and Crowther (13) throughout. We also follow later reporting guidance on common random numbers and on comparative simulation studies (14,15). The study is exploratory. Its rungs and scenarios were not fixed in advance, and the recommendation is the end of a documented sequence of choices rather than a pre-specified comparison. Section 2 describes the simulation design in ADEMP order. Section 3 gives the results rung by rung and then by scenario. Section 4 states the recommendation and its limits.

2 Simulation design

The study simulates complete aggregated N-of-1 trials and analyzes each one under every specification. A single replicate proceeds as follows. Patients are allocated across the randomization paths of the chosen trial design (Section 2.2). For each patient we draw a pre-treatment biomarker together with three latent response components. These are the drug-response component, the natural-history component, and the placebo-response component. The four quantities are drawn jointly across all eight measurement occasions from a multivariate normal distribution. The means follow modified Gompertz trajectories. The covariance matrix contains within-component AR(1) autocorrelation, cross-component correlations, and the biomarker-response correlation that encodes the interaction and depends on treatment state (Section 2.3). The observed symptom score at each occasion is the patient's baseline severity minus the sum of the three components. A beneficial effect therefore lowers the score. Carryover enters through a decay function of time since discontinuation. The decay keeps part of the drug-response component present at nominally off-drug occasions. Every analysis specification in a given cell is then fitted to the same simulated dataset. We record the interaction coefficient, its standard error, and its p-value from each. This is repeated 500 times for every parameter combination in the grid of Section 2.6, 1,000 to 5,000 times in the Hybrid run with independent seeds, and 2,000 times in the supplementary size diagnostic.

Fitting all specifications to a common dataset is deliberate. It makes every contrast among them paired. Paired differences have substantially less Monte Carlo variance than differences between independent draws. Throughout, N denotes the **total** number of patients in a trial. Patients are allocated as evenly as possible across the design's randomization paths, so a design with P paths assigns about N/P patients to each. At $N = 70$ this is 35 per path under the two-path designs and 17 or 18 per path under the four-path Hybrid design.

2.1 Aims

The first aim is to show how each change to the analysis model alters size and power in the biomarker-treatment interaction test of a Hybrid aggregated N-of-1 trial under the correlation encoding. The changes form a ladder of five rungs (Table 3). Rung 0 is the starting analysis. Rung 1 uses a binary drug indicator with a residual AR(1) correlation. Rung 2 adds CR2

Table 1: The six scenario families of the ladder comparisons. Every biomarker is predictive only. Within a cell every analysis is fitted to the same simulated trials; A4 was fitted to the stored trials of families 1 to 5 afterward. Family S has independent seeds for every cell. Section 2.6 describes the cells.

Family	Scenarios	Replicates
1	Base 30-cell grid: three designs, $N \in \{35, 70\}$, $c_{bm} \in \{0, 0.30, 0.45\}$, $t_{1/2} \in \{0, 0.5, 1\}$ weeks	500
2	Misspecified half-life (0.2, 2 or 4 weeks) and decay shape (Weibull $k \in \{0.5, 2\}$), analyzed as exponential	500
3	Serial correlation $\rho \in \{0.5, 0.9\}$	500
4	Patient-level dropout, completely at random or by baseline severity	500
5	Patient slope variance (SD 0.3 per week), Hybrid, $N = 70$	1,000
S	Hybrid run with independent seeds: A3 and A4, $N \in \{35, 70\}$	1,000 to 5,000

cluster-robust inference to fix size. Rung 3 replaces the binary indicator by an exposure-weighted predictor at the true half-life to recover part of the power that carryover removes. Rung 4 chooses the half-life by AIC, on the model without the interaction, to remove the need to know it. Each rung is judged against its predecessor by a paired contrast on the same simulated trials. A side analysis measures what it costs to choose the half-life with the interaction in the model instead.

The second aim is to find where a rung does not help. The scenario families of Table 1 stress the ladder with a misspecified half-life and decay shape, serial correlation, dropout and a patient-specific slope variance. The third aim is to arrive at a recommended analysis for the Hybrid design and to state its conditions and limits. A fourth aim is to document the evidence for each rung so that a reader can check it.

2.2 Trial designs

The three designs differ in how on-drug and off-drug occasions are arranged. They therefore differ in how much post-discontinuation information each generates. All three schedule eight measurement occasions per patient. Each randomization path is a fixed on-drug/off-drug schedule. Each design also carries an expectancy weight, defined as the participant's probability of being on drug by design. The weight is one during open-label periods and one half during blinded periods, and it scales the placebo-response component.

Crossover (CO). A traditional two-period within-patient crossover, blinded throughout. Occasions are spaced 2.5 weeks apart from 2.5 to 20 weeks. One path is active for the first four occasions and then on comparator. The other path reverses that order, so it never discontinues and contributes no post-discontinuation observations.

Open-label followed by blinded discontinuation (OL+BDC). Open-label titration at 4, 8, 12, and 16 weeks, then weekly follow-up to week 20 with a blinded late switch to placebo. The two paths discontinue after week 17 or 18, so only two or three occasions follow.

Hybrid. The reference design of this program, taken from the starting DGP. It consists of

open-label titration at weeks 4 and 8, blinded discontinuation weekly from week 9 to 12, and a brief two-occasion crossover at weeks 16 and 20. The four paths vary discontinuation timing and crossover-arm order. It has the most on/off transitions of the three designs. It is the only one that allows more than one discontinuation per patient. The recommendation of this manuscript is made for this design. CO and OL+BDC are carried through the factorial grid and the corrected grid to show where the recommendation does and does not transfer.

One consequence bears directly on the results. The first post-discontinuation occasion falls one week after switching under OL+BDC and Hybrid, but 2.5 weeks after switching under CO. At the half-lives examined here, roughly half the drug effect persists into the first off-drug occasion under the two non-crossover designs. Less than a fifth persists under CO. The designs therefore differ substantially in how much carryover they can accumulate.

2.3 Data-generating mechanism

Under the correlation encoding, the biomarker and the drug-response component are drawn jointly from a multivariate normal distribution. Their correlation depends on treatment state: $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ is the decay form and t_{sd} is the time since drug discontinuation. The data-generating process has the structure of the recommended DGP of the companion study. Its parameter values are not those of the companion study (Table 2). The response SDs and the biomarker and baseline scales are the simulation core's own set, chosen for this program, not Hendrickson's values. The placebo-response SD in particular is 2 here against 10 in the starting DGP, which makes the placebo pathway much weaker. The structure differs from the starting DGP in four respects. The biomarker-response correlation is zero before first exposure and decays after discontinuation at the carryover half-life. The within-component correlation is a separable AR(1) structure at $\rho = 0.7$. That is, every component has the same first-order autoregressive correlation, and a single cross-component constant $c_1 = 0.2$ is scaled by $\rho^{|w_t - w_s|}$ at every lag. Carryover on the drug-response mean decays from the last on-drug mean. A covariance matrix that is not positive definite stops the simulation rather than being repaired (Section 2.8).

The decay function $\phi : \mathbb{R}_+ \to [0, 1]$ maps t_{sd} to a residual-effect multiplier. Two forms are used. Each is matched to a common half-life $t_{1/2}$. Exponential decay,

$$\phi_{\mathrm{exp}}(t_{sd}) = \exp\left(-\lambda t_{sd}\right), \qquad \lambda = \frac{\ln 2}{t_{1/2}},$$

is classical first-order elimination and the default pharmacokinetic assumption. Weibull decay,

$$\phi_{\mathrm{weib}}(t_{sd}) = \exp\left(-\left(\lambda_w t_{sd}\right)^k\right), \qquad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}},$$

has a shape parameter k. The elimination tail is heavier than exponential when $k < 1$ and lighter when $k > 1$. The form reduces to exponential at $k = 1$. The factorial grid uses $k \in \{0.7, 1.5\}$ alongside exponential. The decay-shape misspecification cells (family 2) use

Table 2: Parameter values of the data-generating process. This paper uses the simulation core's own set ('simcore2026' in 'simulation-core.R'). The companion paper's recommended DGP uses the response and baseline values of the published code with the same correlation structure, which is described in the text. The biomarker enters the analysis centered and scaled, so its location and scale do not affect the interaction test.

Parameter	This paper	Companion paper
Natural history (TV): Gompertz max, rate, displacement	10.99, 0.42, 5	6.51, 0.35, 5
Placebo response (PB): Gompertz max, rate, displacement	6.51, 0.35, 5	6.51, 0.35, 5
Drug response (BR): Gompertz max, rate, displacement	10.99, 0.42, 5	10.99, 0.42, 5
SD of TV, PB, BR	6, 2, 8	10, 10, 8
Biomarker: mean, SD	0, 1	124.3, 15.4
Baseline severity: mean, SD	70, 10	83.1, 18.5
Within-component serial correlation ρ	0.7	0.7
Cross-component constant c_1	0.2	0.2
Biomarker-response correlation c_{bm}	0 to 0.45 by scenario	by scenario

$k \in \{0.5, 2\}$. All forms pass through the same half-life-defining point, $\phi = 0.5$ at $t_{sd} = t_{1/2}$. They are therefore comparable at a fixed nominal half-life even though their shapes differ on either side of it. The decay form the analysis assumes differs by scenario. In the factorial grid the exposure-weighted predictor is built with the data-generating form, shape and half-life. This is the driver's default when no analysis form is given. The Weibull cells of the factorial grid therefore test the representation under a correctly specified decay, not its robustness to the shape. Every other analysis assumes exponential decay. The Weibull cells of family 2 are the shape-misspecification check.

2.4 Estimand and analysis specifications

The target of inference is the on-drug biomarker-response correlation c_{bm}, a dimensionless quantity. Every analysis tests it through the coefficient of $\mathrm{bm} \times X$. Its target on the calibrated scale is $\theta = -c_{bm}\sigma_{BR}$, with $\sigma_{BR} = 8$ in this parameter set (Table 2): the expected difference in on-drug response per SD of the biomarker. Bias, empirical SE and coverage are reported on this scale. Type I error is the rejection rate at $c_{bm} = 0$ and power the rejection rate at $c_{bm} > 0$. What the coefficient estimates depends on X. Under A3, X is exposure-weighted at the true half-life and the coefficient targets θ. Under A1 and A2, X is the binary indicator, which codes residual exposure off drug as none, so under carryover the coefficient differs from θ. Under A4 the half-life is selected, and the coefficient targets θ only when the selected value is the true one. The base analysis model is `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~ t | ptID))`. The analyses differ in how X is constructed and how the standard error is obtained. This manuscript labels them A0 to A4 by rung (Table 3). The table also gives the code each analysis carries in the companion papers and earlier drafts, where A4 is G10, and the `E`-prefixed code under which it is stored in the code and data files described in the Reproducibility section.

Table 3: The five analyses of the ladder. Each rung changes one thing from the one before, except the step from A0 to A1. A3 uses the true half-life and is an oracle benchmark, not a candidate analysis. The companion papers and earlier drafts use the codes in the third column; the data files use the stored codes. Sx is the symptom score, bm the biomarker, Db the binary on-drug indicator, Dbc the exposure-weighted predictor and De the design expectancy. Every analysis has a patient random intercept.

Analysis	Rung	Code in companion papers	Stored code	Formula	Inference	Assumptions
A0	0	E0	E0	Sx ~ bm + t + Db + bm:Db (+ De where expectancy varies); random intercept	lme4 with lmerTest Satterthwaite df, no residual correlation	Binary drug indicator; independent residuals given the random intercept; patients never on drug dropped
A1	1	G1	E1	Sx ~ bm + t + Db + bm:Db; random intercept, corCAR1 residuals	nlme working-covariance standard error	Binary drug indicator; AR(1) residual correlation at the stated working structure
A2	2	G6	E1cr2	Sx ~ bm + t + Db + bm:Db; random intercept, corCAR1 residuals	CR2 cluster-robust standard error, Satterthwaite df	Binary drug indicator; working covariance may be wrong
A3	3	G8	E2cr2	Sx ~ bm + t + Dbc + bm:Dbc; random intercept, corCAR1 residuals	CR2 cluster-robust standard error, Satterthwaite df	Exponential carryover at the true half-life
A4	4	G10	E7null_cr2	A3 formula; half-life chosen by ML-AIC over {0.25, 0.5, 1, 2} weeks on the model without bm:Dbc, then refit by REML with it	CR2 cluster-robust standard error, Satterthwaite df, selection ignored	Exponential carryover; true half-life near the candidate grid; selection does not use the interaction term

2.4.1 The analysis ladder

The ladder orders the analyses by what each one changes. Table 3 gives the formula, the inference and the assumptions of each. Rung 0 is the starting analysis (A0). Rung 1 is the binary-indicator analysis with `corCAR1` residual correlation and the working-covariance standard error (A1). Rung 2 is A1 with CR2 cluster-robust inference (A2). Rung 3 replaces the binary indicator by the exposure-weighted predictor at the true half-life, with CR2 (A3). Rung 4 chooses the half-life by AIC on the model without the interaction, with CR2 (A4). Each rung changes one thing from its predecessor, with one exception. The step from A0 to A1 changes three things at once, and the isolation analyses E0n, E0nc and E1nc separate them (Section 3.2.1). A3 uses the true half-life. No analyst has it, so A3 is an oracle benchmark, not a candidate analysis: it shows what the selection step of rung 4 costs.

2.4.2 Rung 0: the starting analysis (A0)

A0 is the starting analysis, the analysis published with the starting DGP (1). We read it from the published repository (commit `58b32a9`, `R/lme_analysis.R`) rather than from this project's vendored copy, which had acquired a residual correlation structure at some later point. It fits `lme4::lmer` with a random intercept and no residual correlation structure. It uses the binary on-drug indicator and takes Satterthwaite p-values from `lmerTest`. As published it also adds the expectancy main effect D_e whenever expectancy varies within the design, and it drops patients who are never on drug. Under the crossover design expectancy is

constant, so D_e drops out. A0 therefore differs from A1 in three respects at once: no `corCAR1`, the D_e term, and the software and degrees-of-freedom method. A separate simulation changes them one at a time (E0n, E0nc and E1nc; Section 3.2.1).

2.4.3 Rung 1: binary indicator with residual correlation (A1)

Here $X = D_{it}$, the binary on-drug indicator (one on drug, zero off). Carryover is not represented. An off-drug occasion that still carries most of the drug effect is coded the same as one that carries none. This is the default an analyst obtains without considering carryover. We treat it as the benchmark against which the other specifications must justify their additional structure. It uses no half-life, so it does not depend on the analyst's pharmacokinetic assumptions.

2.4.4 Rung 2: CR2 cluster-robust inference (A2 to A4)

The standard error of $\beta_{bm:X}$ read from the fitted `lme` object is valid only if the assumed residual covariance is correct. The working covariance combines a patient random intercept with a `corCAR1` AR(1) residual correlation. The companion calibration study (16) shows that this does not match the covariance the data-generating process induces. We quantify the consequence with the calibration ratio

$$\kappa = \frac{\overline{\widehat{SE}}}{SD(\hat{\beta}_{bm:X})},$$

the mean model-reported standard error across replicates divided by the empirical standard deviation of the replicate point estimates. $\kappa = 1$ is correctly calibrated and $\kappa > 1$ is a conservative test. The mean of the standard errors is below their root mean square. κ as defined is therefore biased downward when the standard errors vary across replicates. We keep the mean because every stored driver output uses it. With the root mean square instead, the family-level κ of the CR2 analyses rises by 0.01 to 0.03, slightly more for A4, whose standard errors vary more with the selected half-life. A2, A3 and A4 are close enough that their order can change, and the alternative κ of A4 reaches 0.95 to 1.02 by family. The empirical SD in the denominator is also sensitive to a heavy-tailed estimate distribution. Section 3.3.2 meets that case under heavy missingness and reports a robust ratio beside it. Coverage of the nominal 95% interval is the complementary measure. It is reported where the stored per-replicate output allows it. The remedy for the mismatch is a cluster-robust sandwich variance. It derives the variance of $\hat{\beta}_{bm:X}$ from observed between-patient variation rather than from the assumed covariance. With only $N = 35$ or 70 clusters the naive Liang-Zeger form is itself biased downward. We therefore use the CR2 bias-reduced estimator of Bell and McCaffrey (8) with Satterthwaite degrees of freedom, as implemented in `clubSandwich` (17). It replaces only the standard error, the reference distribution and the p-value. Point estimates are unchanged. The Kenward-Roger correction (18) is the standard small-sample correction within the working covariance. It corrects the degrees of freedom and the variance under

the working covariance, so it does not address a misspecified one. It is not evaluated here. A2 is A1 with CR2 inference. A3 and A4 also use CR2, so A2 and A3 are the like-for-like comparators for the power of A4. A1 is conservative. Part of any power difference against it is the increase in power from correcting a conservative test.

2.4.5 Rung 3: exposure-weighted predictor (A3)

Here $X = D_{bc,it}$, a continuous exposure-decayed predictor. It is built from the analyst's assumed decay form and analysis half-life $t^a_{1/2}$. At on-drug timepoints $D_{bc,it} = 1$. Off drug after first exposure it decays as $D_{bc,it} = (1/2)^{t_{sd,it}/t^a_{1/2}}$. It loses half its value every $t^a_{1/2}$ weeks after discontinuation. Before any exposure $D_{bc,it} = 0$. This pre-exposure rule was absent in earlier drafts (Section 2.8). The half-life assumption enters only through the predictor values passed to `lme`. Exposure-weighted is therefore a different model for each assumed half-life. As $t^a_{1/2} \to 0$, $D_{bc,it} \to D_{it}$, so Unadjusted is the boundary member of the Exposure-weighted family at zero assumed half-life. At rung 3 the assumed half-life is set equal to the data-generating one. This is the true half-life, which no real analyst knows. Rung 4 removes that assumption. Under the correlation encoding, the regression of the response on the biomarker at an occasion is linear in $c_{bm}\,\phi(t_{sd})$. $D_{bc,it}$ at the true ϕ is therefore the correctly specified regressor. The factorial grid compares a correct model with two incorrect ones. The question it answers is how much the incorrect ones lose.

2.4.6 Rung 4: AIC-selected half-life (A4)

This specification uses the Exposure-weighted formula but chooses the analysis half-life $\hat{t}^a_{1/2}$ from the data. Write $M_0(t^a_{1/2})$ for the `lme` fit of the model without the interaction, Sx $\sim$ $bm + t + D_{bc}$, with $D_{bc,it}$ constructed at a candidate half-life. The selected half-life is

$$\hat{t}^a_{1/2} = \arg\min_{h \in \mathcal{T}} \mathrm{AIC}(M_0(h)), \qquad \mathcal{T} = \{0.25,\ 0.5,\ 1.0,\ 2.0\} \text{ weeks},$$

with $\mathrm{AIC}(M) = -2\log L(M) + 2p$ (19). The candidate fits are compared by maximum likelihood. The candidates differ in a fixed-effects column, and REML likelihoods are not comparable across fixed-effects designs. The model with the interaction is then fitted by REML at $\hat{t}^a_{1/2}$, and the interaction coefficient, its CR2 standard error and its p-value are read from that single fit, not pooled across $\mathcal{T}$. Because the selection model leaves out the interaction, the selected half-life does not depend on the coefficient under test. The test still treats $\hat{t}^a_{1/2}$ as fixed in advance when the same data chose it, so a post-selection risk remains in principle (9–11). It is smaller than when the selection uses the tested coefficient, and Section 3.2.4 measures it against the same predictor at the true half-life (A3). That section also reports the distribution of the selected half-life.

2.4.7 Side analysis: selecting on the interaction model

The obvious alternative makes the same choice with the interaction term in the selection model, Sx $\sim$ $bm + t + D_{bc} + bm{:}D_{bc}$. It mirrors `characterize_carryover()` (`R/carryover_analysis.R`), the package's own real-data half-life-selection routine, and earlier drafts used it as the recommended analysis. The selected half-life then depends on the coefficient under test. Fitted with CR2 to the same simulated trials as A4, it rejected the null more often than A4 in every scenario family, by +0.001 to +0.007, and its standard error was a few percent too small under the alternative. A4 leaves the interaction out of the selection model for this reason. The variant is not reported further.

2.5 Choice of analysis specifications

The specifications were chosen to represent distinct, established positions an analyst can take toward an unmeasured decay, and then to answer the specific questions each rung raised. They were not chosen to survey the wider space of N-of-1 analysis models. Bayesian hierarchical models, GEE as a primary model, randomization tests, flexible spline decay curves, state-space treatments of drug concentration and two-stage meta-analytic combination are all established in the N-of-1 or crossover literature. None is evaluated here. Section 4.4 returns to three of them. An earlier draft also tested carryover terms interacted with the biomarker (`bm:L` and `bm:t_{sd}`). Both had lower power than A1 at every cell. We attribute this to the data-generating process having a single true decay parameter, so that a second interacted coefficient adds estimation noise with no signal to recover. They were dropped in favor of choosing the decay half-life from the data (rung 4), which addresses the same unknown.

2.6 Factorial grid and parameter settings

The factorial grid of job 26 crosses biomarker moderation, carryover half-life, trial design, sample size and decay form. It fits the binary indicator and the exposure-weighted predictor at the true half-life, both with the working-covariance standard error:

Parameter	Values
Biomarker moderation (c_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Sample size (N)	35, 70
Trial design	CO, Hybrid, OL+BDC
Decay form	exponential; Weibull $k \in \{0.7, 1.5\}$
Replicates per cell	500

Three half-lives, three designs, two sample sizes, three biomarker levels and three decay forms give 162 distinct configurations in the correlation-encoding slice reported here. The production grid holds 540 cells, 270 per encoding. Each encoding has the 162 configurations, a Weibull

$k = 1$ condition (54 cells) and a linear decay form (54 cells). The grid also stores a fourth specification (`E4`) from an earlier draft. The Weibull $k = 1$ condition is the exponential form drawn with its own random numbers, so it is a replicate-level duplicate rather than a numerical one. The linear decay form is not pharmacokinetically motivated and is not reported. The `E4` specification is not part of this study. Counts in the text refer to the 162 distinct configurations unless the 216 exponential-plus-Weibull cells are named. The $c_{bm} = 0$ row is the Type I error check against nominal 5%. The $t_{1/2} = 0$ row gives a best-case power reference at each combination of N, design and c_{bm}.

The corrected 30-cell grid (job 27, family 1) fits every specification to the same trial each replicate. The grid uses exponential decay, design $\times$ {CO, Hybrid, OL+BDC}, and $N \in \{35, 70\}$. Panel A varies $c_{bm} \in \{0, 0.30, 0.45\}$ at $t_{1/2} = 1$ week. Panel B varies $t_{1/2} \in \{0, 0.5\}$ at $c_{bm} = 0.45$. The scenario job (family 2) places nine cells on Hybrid at $N = 70$ and $t_{1/2} = 0.5$ week that this paper reports. These are the null and two alternatives ($c_{bm} = 0.45$, 0.25); the OL+BDC null and alternative; a four-fold half-life misspecification (true $t_{1/2} = 2$ weeks, null and $c_{bm} = 0.25$); and a decay-shape misspecification (Weibull $k = 2$ data, exponential analysis, null and $c_{bm} = 0.25$). Its other cells, with a prognostic biomarker or mean moderation, are not reported here.

A supplementary study, specified in a working protocol after those jobs (`32-gap-fill-protocol.md`), then closed the items they left open. All of its cells are on Hybrid with every specification fit to the same trial, at 500 replicates unless stated. It replaces the size diagnostic with four corrected-code null cells at 2,000 replicates ($t_{1/2} \in \{0.5, 1\}$ weeks, $N \in \{35, 70\}$). It extends two scenario cells from one cell to a grid crossing $N \in \{35, 70\}$ with the null and an alternative. The first is Weibull $k \in \{0.5, 2\}$ data analyzed as exponential ($c_{bm} \in \{0, 0.25, 0.45\}$). The second is a true half-life of 0.2, 2 or 4 weeks ($c_{bm} \in \{0, 0.25\}$). The value 0.2 is below the candidate grid of rung 4, 4 is above it, and 2 is at its boundary. The study adds a third scenario, the within-component autocorrelation $\rho \in \{0.5, 0.9\}$, at $c_{bm} \in \{0, 0.30\}$. The largest feasible c_{bm} falls with ρ, and 0.45 is infeasible at $\rho = 0.9$ (the largest value for which the correlation matrix is positive definite is 0.391). Finally, the study imposes monotone dropout at rates of 0.1, 0.2 and 0.3, with $c_{bm} \in \{0, 0.45\}$ and $N \in \{35, 70\}$. Dropout is either completely at random (MCAR) or has a hazard that rises with the baseline severity rank (MAR). The MAR hazard is scaled by twice the rank so that the average hazard is unchanged. The rate is the fraction of patients whose series is incomplete. The first occasion is always observed. The per-occasion hazard over the remaining seven is $1 - (1 - \text{rate})^{1/7}$ (0.015, 0.031 and 0.050), which keeps about 95, 90 and 84 percent of the observations. A first run used the earlier convention of the driver, in which the rate is the hazard at each of the eight occasions including the first. That convention keeps 64, 42 and 27 percent of the observations and leaves 10, 20 and 30 percent of patients with none. It is therefore not a 10 to 30 percent dropout scenario. It is reported separately as a heavy-missingness stress case. The study also ran prognostic cells, which belong to the companion paper (12). Every cell passed a pre-flight check that the smallest eigenvalue of its covariance matrix is positive before any simulation ran.

The Hybrid run with independent seeds (family S) fits A3 and A4 at $t_{1/2} \in \{0.5, 1\}$ week for

the null and $\{0.5, 2\}$ weeks for the alternatives, $N \in \{35, 70\}$, with 5,000 replicates per null cell and 1,000 per alternative. It adds the patient-slope cell of family 5. A4 was later fitted to the stored trials of families 1 to 5 by regenerating each trial from its stored seed; a trial was used only if a refit of a stored analysis reproduced its stored estimate (Reproducibility section).

Sample-size study. A separate run gives the total N that A4 needs for 80 percent power. It fits A4 and A2 to the same simulated trials over a grid of N, for the three designs, $t_{1/2} \in \{0, 0.5, 1\}$ week and $c_{bm} \in \{0.2, 0.3, 0.4\}$, with five values of N per c_{bm} chosen to bracket 80 percent power and 500 replicates per cell. Every cell has its own seed. The run was made under both parameter sets: the set used everywhere else in this paper, and the starting values of commit 58b32a9 that the companion paper uses (Table 2), because the required N depends strongly on the variances. Within a design, half-life and analysis, power is modeled as a probit curve in $c_{bm}\sqrt{N}$, fitted by binomial maximum likelihood over the 15 cells, and inverted at 0.8. The noncentrality of the interaction test is close to proportional to $c_{bm}\sqrt{N}$, so one curve serves all three values of c_{bm}. The interval for N reflects Monte Carlo error under that model only.

2.7 Performance measures and computing environment

For each cell we report Power, Type I error, Bias, Empirical SE (calibrated scale), Coverage of the nominal 95% Wald CI, MSE, non-convergence rate, and the MCSE of each, following Morris et al. (13, Tables 5-6). For the analyses that select the half-life (A4), a fit can fail in the ML selection or in the REML refit at the selected value. The stored output records only whether the final fit succeeded, so the non-convergence rate bounds the refit failure rate from above. Over the 55,000 A4 fits of the backfill, 132 failed (0.24 percent). We chose $n_{sim} = 500$ so that the MCSE on power at $p = 0.5$ would not exceed 0.022. At $n_{sim} = 500$ the two-MCSE band around a nominal 0.05 rejection rate is approximately $[0.031, 0.069]$. That band is a check on gross inflation, not a demonstration of nominal size. A single 500-replicate cell whose true size is 0.065 falls inside the band with probability about 0.64. One whose true size is 0.075 falls inside it with probability about 0.31 (normal approximation). Claims about size therefore rest on the pooled 2,000-replicate cells of Section 3.2.4 and on paired contrasts between specifications. Single-cell band membership is reported as such. No simulation runs while this manuscript renders. All specifications are fit to the same simulated dataset within each cell (common random numbers). Every pairwise contrast between specifications therefore has reduced Monte Carlo variance relative to independent draws. Where a paired standard error is quoted, it is computed from the per-replicate rejection indicators. The drivers also seed replicate r of every cell from a job-level base plus r. Cells at the same design and N within a job therefore share a random-number stream. Contrasts between such cells are paired too. Counts of cells inside a band are not independent confirmations and are not presented as such. Comparisons across jobs, and across the two simulation hosts, are unpaired and carry the full Monte Carlo error of both cells. The two hosts do not produce identical draws. Exact file paths and versions are given in the Reproducibility section.

2.8 Corrections to the simulation code during the study

Two defects in the simulation driver were found and corrected while this study was in progress, and one policy was changed. Each is stated here so that the reader can tell which results it affects.

1. **Pre-exposure coding of the exposure-weighted predictor (corrected 2026-10-06).** Before any exposure, $D_{bc,it}$ was coded as $\phi(0) = 1$ rather than 0. A patient on the placebo-first path of the crossover design was therefore treated as fully exposed at every visit. The same correction replaced the REML AIC comparison of the half-life selection with the ML comparison and REML refit of Section 2.4.6. A paired check fitted each simulated trial under the old and new coding (`26-co-dbc-fix-paired.R`). It showed that the entire collapse of the exposure-weighted family under the crossover design reported in earlier drafts was this defect. With the correction, the exposure-weighted analyses match A1 under CO. Hybrid cells were unaffected, because no patient starts on placebo there.
2. **Cross-factor correlation constant (corrected 2026-10-21).** The documented data-generating process uses a single cross-component correlation $c_1 = 0.2$ scaled by $\rho^{|w_t - w_s|}$ at every lag. The driver had used 0.1 at lags greater than zero. The factorial grid was rerun in full under the corrected constant. Every other comparison was run after it. No result from before this correction is reported. The 3,000-replicate size diagnostic of earlier drafts was replaced by the 2,000-replicate corrected-code diagnostic of Section 3.2.2, which reproduces its conclusion.
3. **Non-positive-definite covariance matrices are refused (2026-10-26).** As in the recommended DGP, the driver stops on such a cell rather than changing it (2). No cell reported here was affected.

The exploratory sensitivity blocks of earlier drafts predate both corrections and are no longer part of this manuscript. Those blocks covered within-component autocorrelation, analyst-truth half-life mismatch, dropout, the effect-size curve, and single-cell comparisons of further specifications and of the CR2 variants. Their material remains in the archived drafts. The autocorrelation, half-life and dropout scenarios were rerun under the corrected code in the supplementary grids of Section 3.3.1. The dropout block of those drafts used the per-occasion convention described in Section 2.6. Under that convention a stated rate of 0.1 to 0.3 removed 36 to 73 percent of the observations. Its rates are therefore not comparable with the patient-level rates reported here.

3 Results

Results follow the ladder. Section 3.1 shows the whole ladder at a glance. Section 3.2 gives the evidence for each rung. Section 3.3 stresses the ladder in scenario families. Section 3.4 reports calibration and coverage. Section 3.5 states which analysis to use. The Monte Carlo standard error on power does not exceed 0.022 at $p = 0.5$ in a 500-replicate cell. Differences

smaller than this should not be over-interpreted. Pooled estimates and paired differences have smaller standard errors and are quoted with them.

3.1 The ladder at a glance

Table 3 defines the analyses. Rungs 0 to 4 form the ladder. Each rung changes one thing from the one before, except the step from A0 to A1, which changes three things at once (Section 3.2.1). A3 uses the true half-life: it is an oracle benchmark, not a candidate analysis, and it shows what the selection step of rung 4 costs.

Figure 1 shows the null rejection rate of each analysis in each scenario family.

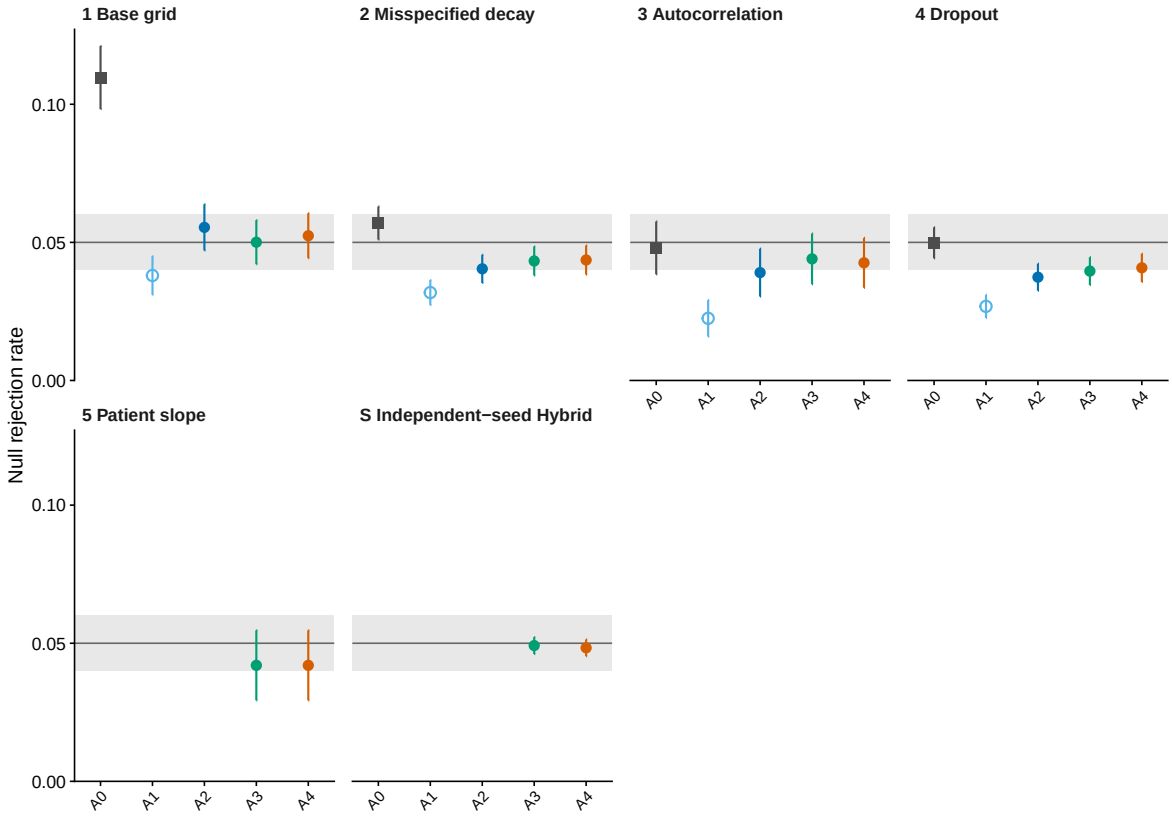


Figure 1: The size ladder. Each panel is one scenario family (Table 1). The vertical axis is the null rejection rate pooled over the null cells of the family, weighted by replicates. The horizontal axis lists the analyses in rung order, A0 to A4. Bars are 2 × the Monte Carlo standard error. The shaded band is 0.04 to 0.06, the two-standard-error band for roughly 2,000 replicates. It is a fixed tolerance around 0.05 and not a test. An analysis that was not fitted in a family has no point.

In the base grid A0 rejects the null at 0.110 (MCSE 0.006), and A1 at 0.038 (0.003). A0 is liberal and A1 is conservative. CR2 (A2) gives 0.055 and exposure weighting at the true half-life (A3) gives 0.050. A4 gives 0.052 (0.004). A4 lies inside the 0.04 to 0.06 band in 6 of 6 families, from 0.041 to 0.052; the oracle A3 does so in 5 of 6. A0 lies outside the band in 1

Table 4: Paired increments along the ladder. Each cell is the later analysis minus the earlier one, with its paired standard error in parentheses, for size (pooled null rejection rate) and power (pooled reference alternative), by scenario family. A dash means the step has no data in that family. The standard errors are conservative because cells within a job share random-number streams. The step from A0 to A1 changes three things at once; its power change is not shown because A0 is liberal.

		Scenario family					
Step	Measure	1	2	3	4	5	S
A0 to A1	Size	-0.072 (0.013)	-0.025 (0.010)	-0.025 (0.009)	-0.023 (0.009)	–	–
	Power	–	–	–	–	–	–
A1 to A2	Size	+0.018 (0.008)	+0.009 (0.006)	+0.017 (0.007)	+0.011 (0.007)	–	–
	Power	+0.024 (0.014)	+0.015 (0.012)	+0.044 (0.012)	+0.016 (0.012)	–	–
A2 to A3	Size	-0.006 (0.007)	+0.003 (0.007)	+0.005 (0.006)	+0.002 (0.006)	–	–
	Power	+0.053 (0.013)	+0.013 (0.012)	+0.010 (0.010)	+0.019 (0.009)	–	–
A3 to A4	Size	+0.002 (0.005)	-0.000 (0.004)	-0.002 (0.004)	+0.001 (0.005)	+0.000 (0.003)	-0.001 (0.002)
	Power	-0.011 (0.008)	+0.003 (0.008)	-0.007 (0.008)	-0.013 (0.007)	-0.004 (0.003)	-0.013 (0.006)

of the 4 families in which it was fitted.

Figure 2 shows power at the reference alternative and the paired change at each step.

A0 is left out of the power comparisons: its test is liberal, so its rejection rate under the alternative is not comparable with that of a correctly sized test. In the base grid power is 0.563 for A1. It rises to 0.587 with CR2 and to 0.640 with exposure weighting at the true half-life. A4 reaches 0.628. Section 3.2.3 gives the design dependence of the gain from exposure weighting.

Table 4 gives the paired increments behind the figures.

In the base grid CR2 changes size by +0.018 (paired SE 0.008) and power by +0.024 (0.014). Exposure weighting changes size by -0.006 (0.007) and power by +0.053 (0.013). Choosing the half-life by AIC on the model without the interaction changes size by +0.002 (0.005) and power by -0.011 (0.008). The rungs therefore fix size at rung 2, recover power at rung 3 and remove the need for the true half-life at rung 4. The patient slope variance is shown in Section 3.3.3.

3.2 Evidence rung by rung

Each subsection opens with what the rung changes and ends with what the rung changed in size and in power. Paired contrasts use the same simulated trials. The numbers in the closing sentences come from Table 4 and the figures of Section 3.1.

3.2.1 Rung 0 to 1: residual correlation

Rung 1 replaces the starting analysis (A0) by A1. A0 fits `lme4::lmer` with a random intercept and takes Satterthwaite p-values from `lmerTest`. It has no residual correlation structure. As published it adds the expectancy main effect D_e whenever expectancy varies within the design, and it drops patients who are never on drug. A1 fits `nlme::lme` with a random intercept and

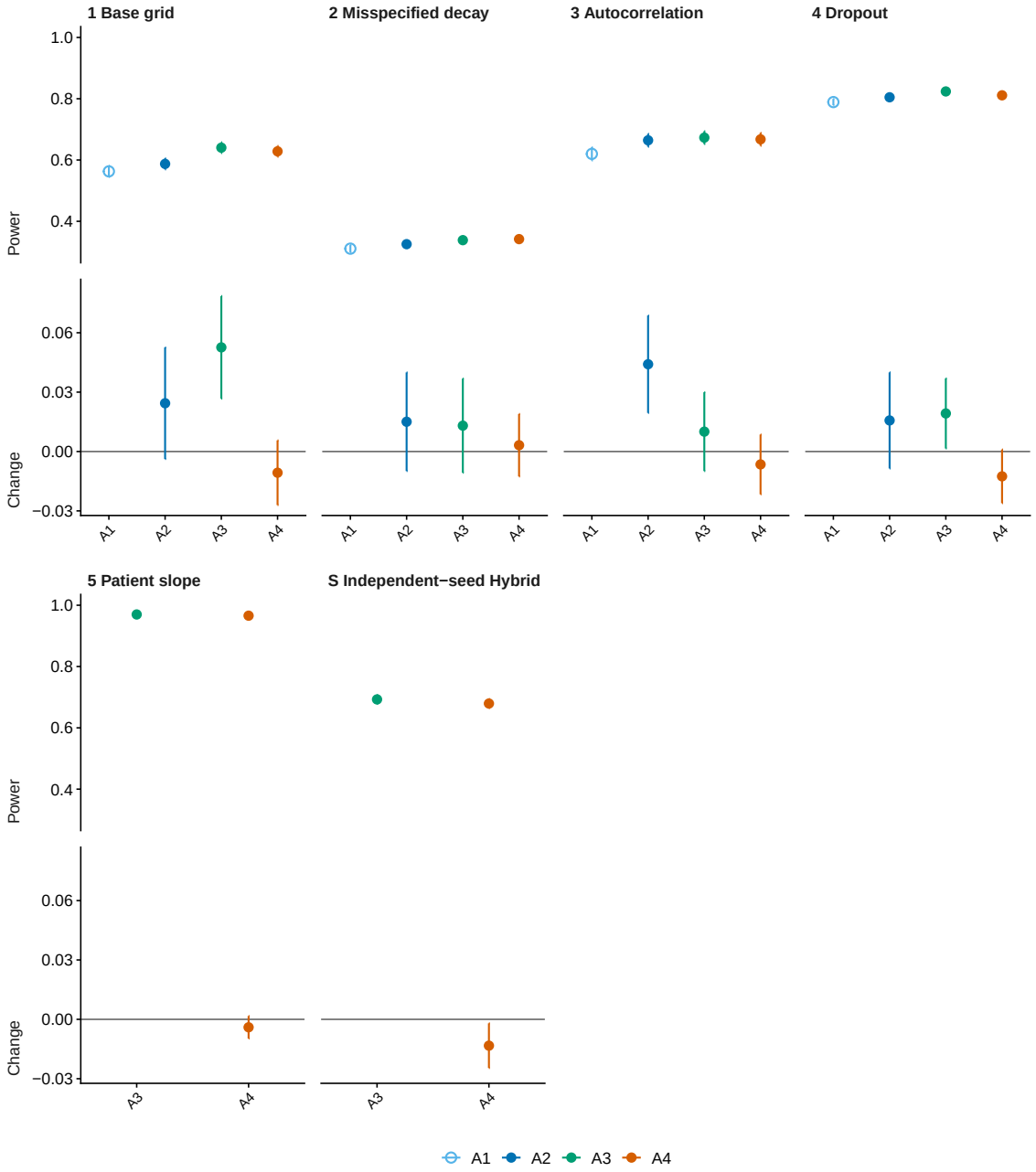


Figure 2: The power ladder. Each panel is one scenario family. A0 is omitted because its test is liberal; its size is in Figure 1. The upper part of a panel is the power at the reference alternative, pooled within the family and weighted by replicates. The lower part is the paired change in power from the preceding rung on the same trials. The horizontal axis is in rung order as in Figure 1. Each row shares one vertical scale across all families, so heights can be compared between panels. The open circle marks A1, which uses the working-covariance standard error; filled circles use CR2. Families 5 and S fitted A3 and A4 only. Bars are 2 $\times$ the Monte Carlo standard error, and the paired standard error for the changes. Paired standard errors are conservative because cells within a job share random-number streams.

a `corCAR1` residual correlation. It has no expectancy term, keeps every patient, and uses the working-covariance standard error. A0 and A1 therefore differ in three respects at once. These are the software and degrees-of-freedom method, the residual correlation, and the expectancy term with the exclusion of patients. Table S1 reports Type I error at the six null cells of the corrected grid. The grid has three designs and two sample sizes, all at $t_{1/2} = 1$ week. It has no null cell at other half-lives outside Hybrid.

A0 rejects at 0.060 to 0.150 in these six cells. It is far outside the two-MCSE band under CO and OL+BDC and at its edge under Hybrid. A1 rejects at 0.030 to 0.048, below nominal at every cell. The step from A0 to A1 therefore moves the test from liberal to conservative.

A0 is liberal under CO and OL+BDC, and at the edge of the band under Hybrid. Across the four Hybrid and OL+BDC null scenarios of Table S7, A0 rejects at 0.060 to 0.100. It reaches 0.100 under OL+BDC. In the 2,000-replicate Hybrid cells of Table S3 it is 0.062 to 0.064. Taken together, its inflation is clear under CO and OL+BDC and marginal under Hybrid, where several single cells lie inside the 500-replicate band. The power of A0 at the alternatives is not comparable with that of the other analyses because the test is liberal.

A separate run attributes the change from A0 to A1 (simrun job 42). It fits five analyses to the same simulated trials and changes one element at a time. E0n refits the formula and the sample of A0 with `nlme` and no residual correlation. It changes only the software and the degrees-of-freedom method. E0nc adds `corCAR1` to E0n. E1nc drops the expectancy term and the exclusion rule of A0 and keeps `nlme` without residual correlation. A1 adds `corCAR1` to E1nc. The run covers the 30 cells of the grid at 1,000 replicates per cell. E0nc failed to converge in at most 9 replicates of a cell. The other four analyses converged in every replicate. Table S2 gives the results at $t_{1/2} = 1$ week.

The residual correlation carries the whole size correction. Adding `corCAR1` to E0n lowers Type I error from 0.137 (MCSE 0.008) to 0.047 (0.005) under CO, from 0.098 (0.007) to 0.034 (0.004) under OL+BDC, and from 0.061 (0.005) to 0.034 (0.004) under Hybrid. Pooled over the six null cells the paired change is -0.061 (paired SE 0.008). Adding `corCAR1` to the A1 formula (E1nc to A1) has the same effect, -0.060 (0.008). Without the residual correlation the standard error is too small. The calibration ratio κ of E0n is 0.75 to 0.97 at the null cells. With `corCAR1` it is 0.99 to 1.08. The overcorrection below nominal under Hybrid and OL+BDC is the conservatism that rung 2 removes.

The software and the degrees-of-freedom method have no effect. A0 and E0n give p-values that differ by less than 0.0001 in every replicate. Their rejection rates agree in all 30 cells.

The expectancy term and the exclusion rule have almost no effect. The exclusion rule of A0 drops no patient in any replicate of this grid, because the drug indicator varies within patients in every design. Under CO the expectancy term is absent from A0, so E0n and E1nc are the same model. Under Hybrid and OL+BDC, removing the expectancy term changes Type I error by -0.005 to +0.001 at the null cells (paired SE at most 0.003). It changes power by -0.015 to +0.000 at the alternative cells (paired SE at most 0.004). A0 is therefore liberal because it has no residual correlation structure.

The residual correlation also carries the change in power. Pooled over the six reference alternatives, adding `corCAR1` to E0n changes power by -0.124 (paired SE 0.013), and adding it to E1nc by -0.129 (0.013). At $t_{1/2} = 1$ week it lowers power in all 12 alternative cells, by 0.040 to 0.193. Part of this loss is the removal of the excess rejections of a liberal test. There are 4 exceptions, all at $t_{1/2} = 0$ under Hybrid and OL+BDC. There `corCAR1` raises power by 0.007 to 0.042 (paired SE 0.003 to 0.013). The gain exceeds two paired SE in 3 of these cells. The residual correlation makes the estimate more precise in these designs. Under Hybrid at $N = 35$ and $t_{1/2} = 0$ the empirical standard deviation of the estimate falls from 1.273 to 1.086. Under CO the empirical standard deviation changes by at most 4 percent, and the reported standard error rises by 28 to 29 percent. Under CO the residual correlation therefore corrects only the standard error.

The residual correlation changes the estimate under carryover. At $t_{1/2} = 1$ week and $c_{bm} > 0$, adding `corCAR1` moves the mean estimate toward zero by 12 to 16 percent under Hybrid and OL+BDC and by 2 to 4 percent under CO. At $t_{1/2} = 0$ the change is at most 1 percent. The binary indicator does not represent carryover, so its coefficient depends on how the fit weights the off-drug occasions. A plausible mechanism is that generalized least squares under a serial correlation gives less weight to an occasion that is strongly correlated with its neighbors. The first off-drug occasions sit next to on-drug ones, and they carry most of the residual effect. We did not verify this mechanism.

Rung 1 therefore turns a liberal test into a conservative one. In the base grid A1 rejects the null at 0.038 against 0.110 for A0. It also costs power. A1 has 0.563 against 0.691 for A0 (Figure 2). The comparison of power is not like for like, because A0 rejects too often under the null. Rungs 2 to 4 recover power with a correctly sized test.

3.2.2 Rung 2: CR2 cluster-robust inference

Rung 2 replaces the working-covariance standard error of A1 by a CR2 cluster-robust standard error with Satterthwaite degrees of freedom (A1 to A2). The point estimates do not change. Across the null cells of the factorial grid the working-covariance test rejects at about 0.037 on average (Section 3.2.3 gives the range). We asked whether CR2 corrects it without disturbing the ranking of the analyses. Table S3 diagnoses the conservatism at four Hybrid null cells with 2,000 replicates each. That count is enough to resolve the calibration ratio κ of Section 2.4.4 (two-MCSE band around 0.05: $[0.040, 0.060]$).

The working-covariance test is conservative for both regressors at every cell. It is most conservative with the exposure-weighted predictor (κ 1.05 to 1.07, Type I error 0.033 to 0.037). The CR2 standard error brings κ to within 0.10 of one and Type I error inside the band for A2, A3 and A4 at all four cells. A4 rejects at 0.048 to 0.053 with κ 0.96 to 0.99, against 0.97 to 0.99 for A3. A0 rejects at 0.062 to 0.064 at all four cells.

Table S1 also reports Type I error at the six null cells of the base grid. A1 is at 0.03 to 0.05 throughout. A2, A3 and A4 are at 0.04 to 0.07, inside the two-MCSE band at every cell except A2 at OL+BDC, $N = 35$ (0.074, at its edge). A separate check with 5,000 independent null datasets per cell, under this paper's parameter set, gave A2 0.049 (MCSE 0.003) at $N = 35$ and

0.056 (MCSE 0.003) at $N = 70$ in OL+BDC (`analysis/scripts/cr2-olbdc-check/`, output in `analysis/data/cr2-olbdc-check/`). The 0.074 is therefore a Monte Carlo excursion of a 500-replicate cell, not a small-sample failure of CR2. The rate at $N = 70$ is about two MCSE above 0.05, a small excess. The median Satterthwaite degrees of freedom of the CR2 test were 12 at $N = 35$ and 24 at $N = 70$. So few degrees of freedom widen the reference distribution. That is part of the power cost of CR2 at small N. A0, the starting analysis, rejects at 0.06 to 0.15. It is far outside the band under CO and OL+BDC and at its edge under Hybrid.

Figures 3 and 4 show power for the five analyses on the corrected grid (job 27). The layout follows Figure 4 of Hendrickson et al. Panel A varies the biomarker effect and Panel B varies the carryover half-life.

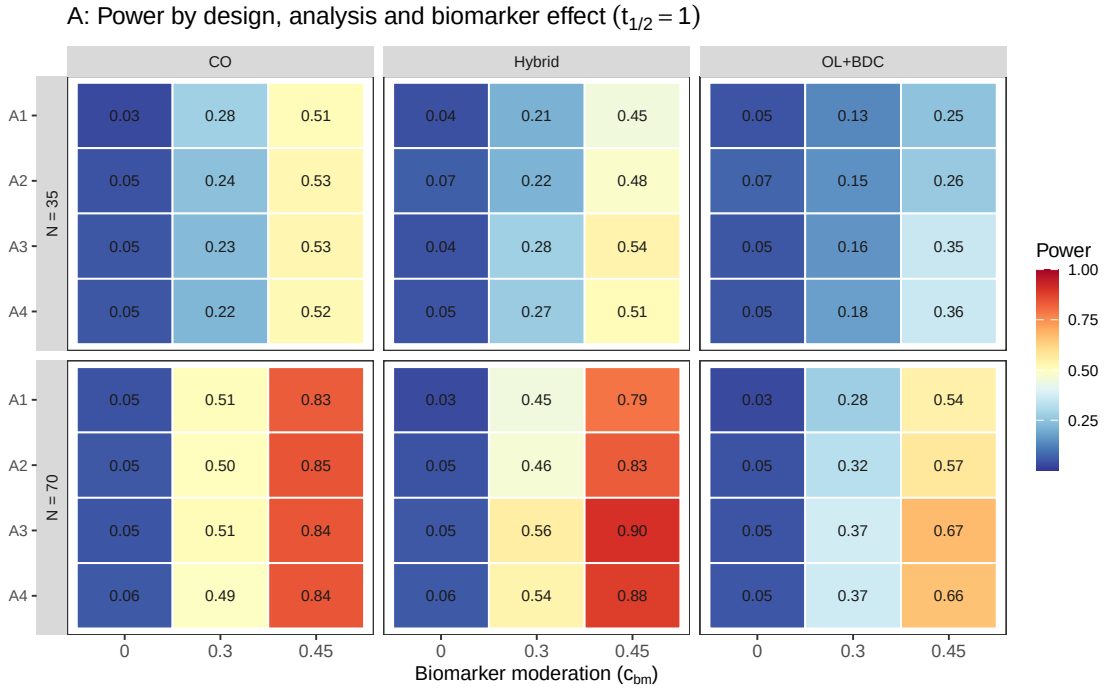


Figure 3: Power by trial design (columns), N (row blocks) and analysis (rows within block) across $c_{bm} \in \{0, 0.30, 0.45\}$ at $t_{1/2} = 1.0$ weeks, exponential DGP, corrected simulation code ($n_{sim} = 500$). The $c_{bm} = 0$ column is Type I error. A4 was fitted afterward to the same simulated trials. A0 is omitted because its test is liberal; its size is in Figure 1 and Table S1.

CR2 changes power as well as size, and not in the same direction everywhere. At the Hybrid reference cell A3 reaches 0.904 against 0.886 for the same predictor with the working-covariance standard error. Over the alternative cells of the grid, adding CR2 to the binary indicator or to the exposure-weighted predictor raises power by a mean of 2.4 points under Hybrid and 3.3 under OL+BDC. Under CO it changes power by a mean of -1.0 points (minimum -5.0). The gain under the two designs with a discontinuation phase is the increase in power from correcting a conservative test. A3 and A4 lead, within a few points of each other, at the Hybrid and OL+BDC alternative cells with carryover. A2 tracks A1 below them. Under CO, A1 to A4 are within a few points of one another. CR2 therefore fixes size without disturbing the ranking. In what follows A2 and A3 are the like-for-like power comparators for A4. A1 is

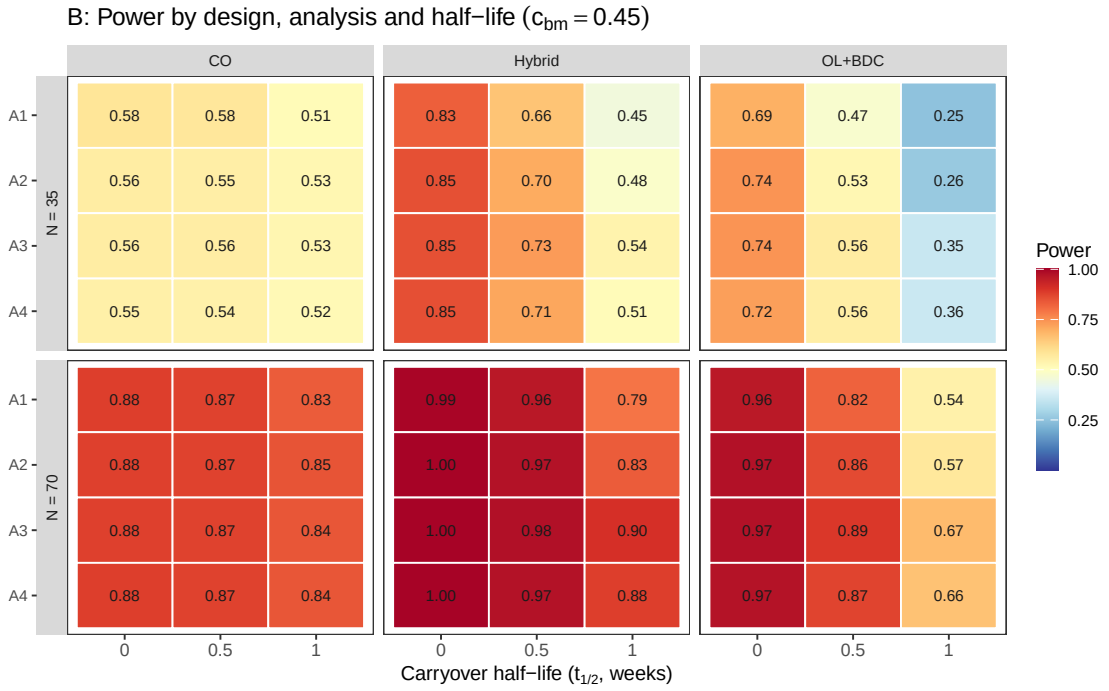


Figure 4: Power across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks at $c_{bm} = 0.45$, exponential DGP, corrected simulation code ($n_{sim} = 500$).

retained because it is what an analyst who does not use CR2 would run.

Rung 2 therefore fixes size. In the base grid CR2 moves the null rejection rate from 0.038 to 0.055, a paired change of +0.018 (SE 0.008). Power changes by +0.024 (SE 0.014). The regressor is unchanged, so the gain comes only from correcting a conservative test. Figure 1 and Table 4 show the same step in the other scenario families. The rung does not help under the crossover design, where CR2 changes power by a mean of -1.0 points.

3.2.3 Rung 3: exposure weighting

Rung 3 replaces the binary on-drug indicator by the exposure-weighted predictor at the true half-life. With CR2 inference the step is A2 to A3. We asked how much of the power that carryover removes the predictor recovers. Table 5 and Figure 5 come from the corrected 30-cell grid (job 27, 500 replicates, every analysis fitted to the same trials). The decay-form comparison and the relative losses over the full factorial come from job 26 (216 cells, 500 replicates, including Weibull decay shapes 0.7 and 1.5 and $N = 35$). Each statement below names its job.

Exposure weighting recovers part of the loss. In the Hybrid reference cell, power under A3 falls from 0.996 with no carryover to 0.976 at $t_{1/2} = 0.5$ and 0.904 at $t_{1/2} = 1$ week. A2 gives 0.831 at $t_{1/2} = 1$. The advantage of A3 is 7.2 points. Its paired standard error is 0.013 (39 replicates rejected by A3 only against 3 by A2 only). With the working-covariance standard error the exposure-weighted predictor leads A1 by 9.6 points (paired SE 0.014). The

Table 5: Reference cell on the corrected 30-cell grid (job 27: Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential decay, 500 replicates, every analysis fitted to the same trials; A4 fitted afterward to the same trials). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$ with its Monte Carlo standard error. Type I is the rejection rate in the null cell of the same design and N. That cell exists only at $t_{1/2} = 1$ week. Empirical SD is the standard deviation of the point estimates and Mean SE the mean reported standard error. κ is their ratio. Bias is the mean estimate minus $\theta = -c_{bm} \cdot 8 = -3.6$ and is shown only for the exposure-weighted analyses A3 and A4. A1 and A2 estimate the coefficient of the binary indicator, which is a different quantity. A0 is shown for size only, because its test is liberal and its power is not comparable. A1 uses the working-covariance standard error; A2, A3 and A4 use CR2. Non-conv is the fraction of fits that did not converge or gave no p-value.

$t_{1/2}$	Analysis	Power	Type I	Empirical SD	Mean SE	κ	Bias	Non-conv
0.0	A0	–	–	–	–	–	–	0.000
	A1	0.992 (0.004)	–	0.73	0.88	1.21	–	0.000
	A2	0.996 (0.003)	–	0.73	0.71	0.98	–	0.004
	A3	0.996 (0.003)	–	0.73	0.71	0.98	-0.03	0.004
	A4	0.996 (0.003)	–	0.75	0.74	0.98	-0.16	0.000
0.5	A0	–	–	–	–	–	–	0.000
	A1	0.960 (0.009)	–	0.76	0.88	1.16	–	0.000
	A2	0.972 (0.007)	–	0.76	0.75	0.99	–	0.002
	A3	0.976 (0.007)	–	0.86	0.85	0.99	-0.02	0.006
	A4	0.972 (0.007)	–	0.86	0.81	0.94	+0.21	0.006
1.0	A0	–	0.060 (0.011)	–	–	–	–	0.000
	A1	0.790 (0.018)	0.032 (0.008)	0.79	0.89	1.12	–	0.000
	A2	0.831 (0.017)	0.046 (0.009)	0.79	0.78	0.99	–	0.004
	A3	0.904 (0.013)	0.054 (0.010)	1.05	1.05	1.00	-0.02	0.000
	A4	0.881 (0.015)	0.056 (0.010)	1.28	1.05	0.82	+0.06	0.006

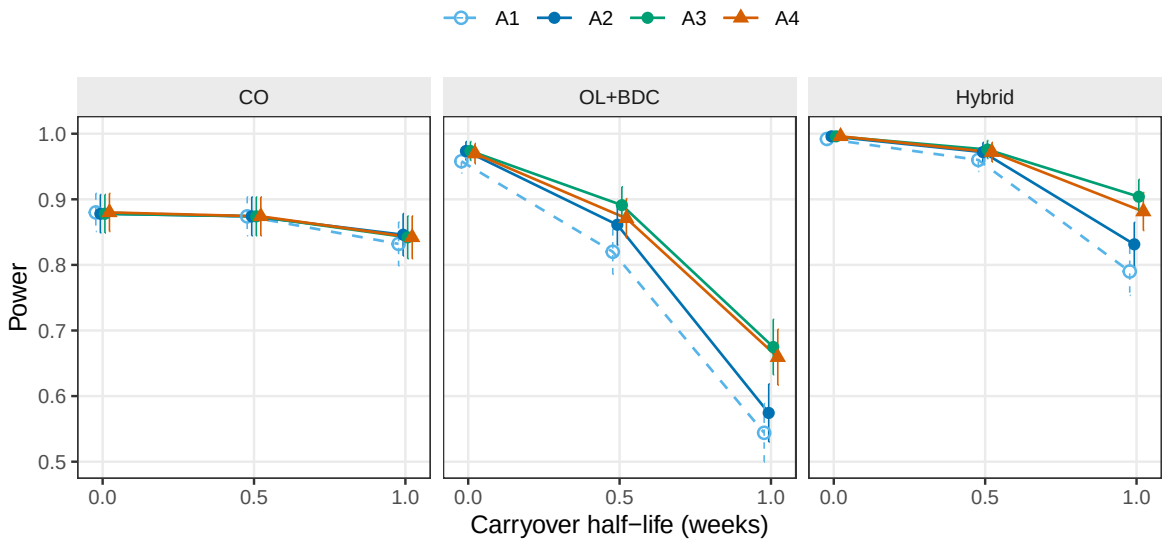


Figure 5: Power against carryover half-life by analysis and design (job 27: $N = 70$, $c_{bm} = 0.45$, exponential decay, 500 replicates). A0 is omitted because its test is liberal. The analyses are A1 (binary indicator, working-covariance standard error); A2 (binary indicator, CR2); A3 (exposure-weighted at the true half-life, CR2); and A4 (exposure-weighted at the half-life chosen by AIC without the interaction, CR2), fitted afterward to the same trials. Points at each half-life are offset horizontally for legibility. Bars are 2 × the Monte Carlo standard error.

advantage is zero at $t_{1/2} = 0$, where there is no carryover to model, and grows with the half-life (Figure 5). Job 26 gives the relative loss of A1 from $t_{1/2} = 0$ to 1 week across N and c_{bm}. It is 47% to 66% under OL+BDC, 19% to 56% under Hybrid and 7% to 16% under CO. At 0.5 week the losses are 16% to 34%, 5% to 25% and 0% to 5%. Carryover therefore removes up to half or more of the power under the two designs with a blinded-discontinuation phase. Exposure weighting buys back only part of it. In job 27 at $c_{bm} = 0.45$, A3 recovers 15% to 44% of A2's loss from $t_{1/2} = 0$ to 1 week under Hybrid and 19% to 25% under OL+BDC, at $N = 35$ and 70. A3 is the oracle: it uses the true half-life and the true decay form. Most of the loss is therefore information the off-drug visits no longer carry, not something a better analysis can recover. A planner who expects a long half-life has to answer it with the design, for example with more participants. We did not simulate design changes such as more off-drug visits.

The advantage depends on the design. At $N = 70$, $c_{bm} = 0.45$ and $t_{1/2} = 1$ in job 27, A3 leads A2 by 10 points under OL+BDC (0.675 against 0.574). OL+BDC has the fewest post-discontinuation occasions, so it has the most to gain from weighting them. A3 leads A2 by 7 points under Hybrid. Under the crossover design the two are within 0.004 of each other (0.842 against 0.846). The first off-drug occasion under CO falls 2.5 weeks after discontinuation, where less than a fifth of the effect persists, so there is little residual exposure to represent. In the decay-form factorial of job 26, over the 108 alternative configurations, the exposure-weighted predictor (working-covariance standard error) is below A1 in 4, by at

most 0.006. That difference is within the noise of a 500-replicate cell. The Weibull $k = 1$ cells of job 26 repeat the exponential configurations with fresh draws. They differ from their exponential counterparts by up to 0.048 in power, which gives the scale of that noise. Earlier drafts of this manuscript reported the exposure-weighted predictor as markedly inferior to the binary indicator under CO. That result was the pre-exposure coding defect of Section 2.8 and does not survive its correction.

The working-covariance tests are conservative on average at every rung. Under the null of job 26 the working-covariance test rejects at a mean of 0.037 across the exponential null cells (range 0.016 to 0.056) against nominal 0.05. The conservatism is the same for both regressors, so it does not disturb the comparison between them. Rung 2 addresses it. In the reference cell the calibration ratio of the exposure-weighted predictor is 1.15 with the working-covariance standard error against 1.00 with CR2 (A3) at $t_{1/2} = 1$ (Table 5).

Rung 3 therefore recovers power and leaves size unchanged. In the base grid A3 gains +0.053 (paired SE 0.013) over A2 and changes size by -0.006 (0.007; Figure 2, Table 4). The rung does not help under the crossover design. It assumes the true half-life, which no real analyst knows. Rung 4 removes that assumption.

3.2.4 Rung 4: choosing the half-life

Rung 4 replaces the true half-life by the half-life that AIC selects from the data, on the model without the interaction (Section 2.4.6). We asked two questions. The first is whether the selection recovers the power that the predictor obtains at the true half-life. The second is what the selection step costs in size and in standard-error calibration. A4 was fitted afterward to the stored trials of families 1 to 5, so every comparison with A3 below is paired on the same trials.

Power. In the base grid A4 changes power against A3 by -0.011 (paired SE 0.008). Over the families of Table 1 the change is -0.013 to +0.003. Table S4 shows that the selection does not recover the half-life. At a true value of 0.5 week the smallest candidate is chosen in about 62 to 63 percent of replicates and the true value in about 27 to 30 percent. At 1 week the choice is spread over all four candidates. The small power cost therefore reflects the insensitivity of the exposure weight to the half-life over this range, not recovery of it.

Size. At the four 2,000-replicate Hybrid null cells of Table S3, A4 rejects at 0.048 to 0.053 against 0.046 to 0.052 for A3. Pooled by family, the change in size from A3 to A4 is -0.002 to +0.002. The Hybrid run with independent seeds (family S) gives the sharpest check. It fitted A3 and A4 to the same simulated trials in four Hybrid null cells at 5,000 replicates each, with an independent random-number stream for every cell (Table S5), and in eight alternative cells at 1,000 replicates each (Table S6). Under the null A4 differs from A3 by -0.0012 to -0.0002, within one paired standard error in every cell. Its calibration ratio there is 0.94 to 0.99. At the alternatives A4 is below A3 by 0.006 to 0.022 in power. Its calibration ratio there is 0.87 to 0.99, against 0.94 to 1.02 for A3. The selection step therefore costs no size but leaves the standard error of A4 a few percent too small under the alternative (Section 3.4).

Rung 4 therefore removes the need to know the half-life. A4 is the leading analysis of the ladder among those that do not need the true half-life: its size is that of the oracle within Monte Carlo error, and its power is within about one point of the oracle's pooled by family.

3.3 Stress scenarios

The ladder was built on the base grid. A real trial departs from that grid in several ways. Families 2 to 5 and S of Table 1 stress the ladder with a misspecified half-life and decay shape, serial correlation, dropout and a patient-specific slope variance. Each subsection states what was simulated and what each rung did. A rung that does not help is named as such.

3.3.1 Misspecified half-life, decay shape and serial correlation

The scenario job placed one cell each at a four-fold half-life misspecification and a Weibull decay analyzed as exponential, beside Hybrid and OL+BDC reference cells. Table S7 gives Type I error at its null scenarios and Table S8 power at its alternatives. A4 was fitted afterward to the four misspecification cells only.

The largest differences are at the half-life misspecification cell, where the true $t_{1/2} = 2$ weeks. A4 gives 0.253, against 0.180 for the binary indicator A1. Under a Weibull $k = 2$ data-generating decay analyzed as exponential, A4 gives 0.661, so the advantage of exposure weighting at rung 3 is not specific to the exponential case at this effect size.

The supplementary study extended these two scenarios from one cell to a grid crossing $N \in \{35, 70\}$ with the null and an alternative. It added the within-component autocorrelation ρ as a third scenario (Section 2.6). Table S9 gives Type I error and Table S10 gives power.

The A4 Type I error lies inside the band at all fourteen null cells, from 0.032 to 0.052. These cells share random-number streams within a job and are 500-replicate cells. The result is therefore consistent with the 2,000-replicate cells of Table S3 but is not fourteen independent confirmations. The working-covariance tests, A1 and the exposure-weighted predictor with the same standard error, are conservative throughout. They are most conservative at $\rho = 0.9$, where the working covariance is furthest from the data. There the Type I error is 0.016 to 0.018, and κ is 1.13 to 1.18 at the null and up to 1.29 at the alternative. CR2 corrects that fully, and A0 is not liberal there. At the 18 alternative cells of the three grids, A4 is at or above its equal-size comparator A2 at all but 5 (smallest margin -0.010). It is within 0.021 of A3 at the true half-life. It is at or above the working-covariance A1 at all but 3 (smallest margin -0.010). The largest margins over A1 occur where the true half-life is long (0.234 against 0.176 for A1 at 2 weeks, $N = 70$) and where the autocorrelation is strong (0.804 against 0.684 for A1 at $\rho = 0.9$, $N = 35$). The Weibull $k = 2$ alternatives at $c_{bm} = 0.25$ are in the summary file but not in the printed table, which shows one alternative per scenario.

Table 6: Type I error and power under monotone dropout (Hybrid, $t_{1/2} = 0.5$ week; the rate is the fraction of patients with an incomplete series; $n_{sim} = 500$; band $[0.031, 0.069]$). A4 was fitted afterward to the same trials. A0 has no power column because its test is liberal.

	Type I error					Power, $c_{bm} = 0.45$			
Dropout (rate, N)	A0	A1	A2	A3	A4	A1	A2	A3	A4
MAR 0.1, $N = 35$	0.040	0.028	0.038	0.042	0.044	0.672	0.683	0.719	0.697
MAR 0.2, $N = 35$	0.052	0.034	0.042	0.046	0.046	0.638	0.651	0.687	0.669
MAR 0.3, $N = 35$	0.054	0.030	0.046	0.044	0.048	0.596	0.593	0.630	0.593
MCAR 0.1, $N = 35$	0.044	0.030	0.042	0.048	0.054	0.674	0.697	0.716	0.700
MCAR 0.2, $N = 35$	0.046	0.024	0.038	0.040	0.042	0.644	0.663	0.686	0.663
MCAR 0.3, $N = 35$	0.044	0.024	0.034	0.036	0.036	0.608	0.620	0.646	0.623
MAR 0.1, $N = 70$	0.058	0.026	0.038	0.040	0.038	0.956	0.970	0.980	0.974
MAR 0.2, $N = 70$	0.056	0.022	0.038	0.038	0.040	0.936	0.964	0.972	0.974
MAR 0.3, $N = 70$	0.056	0.038	0.038	0.040	0.038	0.910	0.940	0.940	0.946
MCAR 0.1, $N = 70$	0.056	0.022	0.032	0.036	0.038	0.958	0.970	0.978	0.978
MCAR 0.2, $N = 70$	0.052	0.020	0.030	0.032	0.026	0.950	0.964	0.976	0.970
MCAR 0.3, $N = 70$	0.040	0.024	0.032	0.032	0.038	0.926	0.942	0.956	0.946

3.3.2 Dropout

The scenarios so far use complete data. The last block of the supplementary study imposes monotone dropout (Section 2.6) on 10, 20 and 30 percent of patients. Dropout is either completely at random or a function of the baseline severity rank. The block runs at $N \in \{35, 70\}$ and 500 replicates (Table 6).

The A4 Type I error is 0.026 to 0.054. It is inside the band at 11 of the 12 null cells. No cell is above the band for A4 or for the starting analysis A0 (0.040 to 0.058). The working-covariance A1 is below the band at 10 of the 12 cells (0.020 to 0.038), as it was without dropout. At the alternative A4 is more powerful than A1 at 11 of 12 cells, by -0.003 to 0.038. Dropout costs power mainly at $N = 35$. Within the dropout block, A4 falls from 0.697 to 0.700 at rate 0.1 to 0.593 to 0.623 at rate 0.3. At $N = 70$ it falls from 0.974 to 0.978 to 0.946 to 0.946. The block has no cell without dropout, so the loss from no dropout to rate 0.1 is not measured on the same trials. The earlier convention of the driver, in which the rate is the per-occasion hazard, removes far more data (Section 2.6). It gives the heavy-missingness stress case. Its cells are not tabulated here. They are in `g9-gap-fill/job-31/summary.rds`. A4 was not fitted to them, so A3 is reported. The A3 Type I error is 0.022 to 0.064. It is above the band at 0 of 12 cells. Its κ falls to 0.60. Power falls steeply for every analysis. A3 falls to 0.085 at MCAR rate 0.3 and $N = 35$ and to 0.214 at $N = 70$. A3 exceeds A1 at 7 of 12 cells. At that level of missingness the retention of patients matters more than the choice of specification. The low κ at $N = 35$ is not a calibration failure of the kind seen under the alternative above. The estimate distribution in that cell is heavy-tailed, so the empirical SD in the denominator is dominated by a few extreme fits. The Type I error in the same cell is below nominal. A robust ratio (median standard error to a MAD-based scale) is much closer to one. Table S11 gives the figures.

3.3.3 Patient slope variance

The referee-variant runs added a patient-specific linear slope to the natural-history component, with standard deviation 0.3 outcome units per week and unrelated to the biomarker. It is a heterogeneity stress, not a prognostic one. A3 was fitted in the run, and A4 was fitted to the same trials afterward, at $N = 70$ and 1,000 replicates (Table S12).

The slope variance changes neither size nor power. A4 rejects at 0.042 under the null and 0.966 at $c_{bm} = 0.45$. Without the slope variance, the same cell of the independent Hybrid run gives 0.051 and 0.960. We also refitted A4 to the same trials with a random slope on time, in both the half-life selection and the final fit. Size was 0.041 and power 0.968. The two fits reached the same decision in all 1,985 replicates in which both converged. The random slope added failures of the final fit (13 of 2,000 against 2); no half-life selection failed. At this slope variance a random slope on time neither gains power nor changes size.

3.4 Calibration and coverage

The calibration ratio κ (Section 2.4.4) and the coverage of the nominal 95 percent interval measure whether the reported standard error describes the spread of the estimates. Figure 6 gives κ by scenario family and analysis, and the coverage of A3 and A4 where the per-replicate output allows it.

A1, and the exposure-weighted predictor with the same working-covariance standard error, have κ above one in every family. The range is 1.08 to 1.17 for A1 and 1.09 to 1.17 for the exposure-weighted predictor. CR2 brings A2 and A3 to 0.96 to 1.02 and 0.97 to 1.02. A4 is from 0.93 to 1.01. The shortfall belongs to the selection step and not to CR2: the standard error of A4 is too small. Pooled by family the shortfall is a few percent. In single cells it is larger. In the independent Hybrid run the calibration ratio of A4 at the alternative falls to 0.87, and at the reference cell of Table 5 it is 0.82. The coverage of the A3 interval over families 2 to 4 is 0.948 to 0.959, and that of the A4 interval 0.940 to 0.948.

The standard error of A4 is close to calibrated at the null (κ 0.99 to 1.06 in the dropout block). It is somewhat too small at the alternative (κ 0.93 to 0.99). That shortfall is not a dropout effect. Across every alternative cell of the supplementary study without heavy missingness, the mean κ of A4 is 0.97 at $N = 35$ and 0.99 at $N = 70$ (minimum 0.92). A3, the same predictor and inference at the true half-life, has 1.02 and 1.02. A2 has 1.00 and 1.03. The CR2 standard error without selection is calibrated under the alternative. The shortfall is specific to the data-chosen half-life and is present with complete data (Section 3.2.4). A nominal 95% interval for A4 will therefore cover below its nominal rate under the alternative at these sample sizes. Table 7 quantifies the amount where the per-replicate output allows it.

The A3 interval covers at its nominal rate. The A4 interval covers at 0.946 to 0.955 at $N = 35$ and 0.934 to 0.941 at $N = 70$, with a minimum lower bound of 0.869. It is close to nominal at $N = 35$ and about one point short at $N = 70$. Selecting the half-life without the term under test therefore protects the test better than the interval. A selection-aware interval, for example a parametric bootstrap that repeats the selection, has not been evaluated.

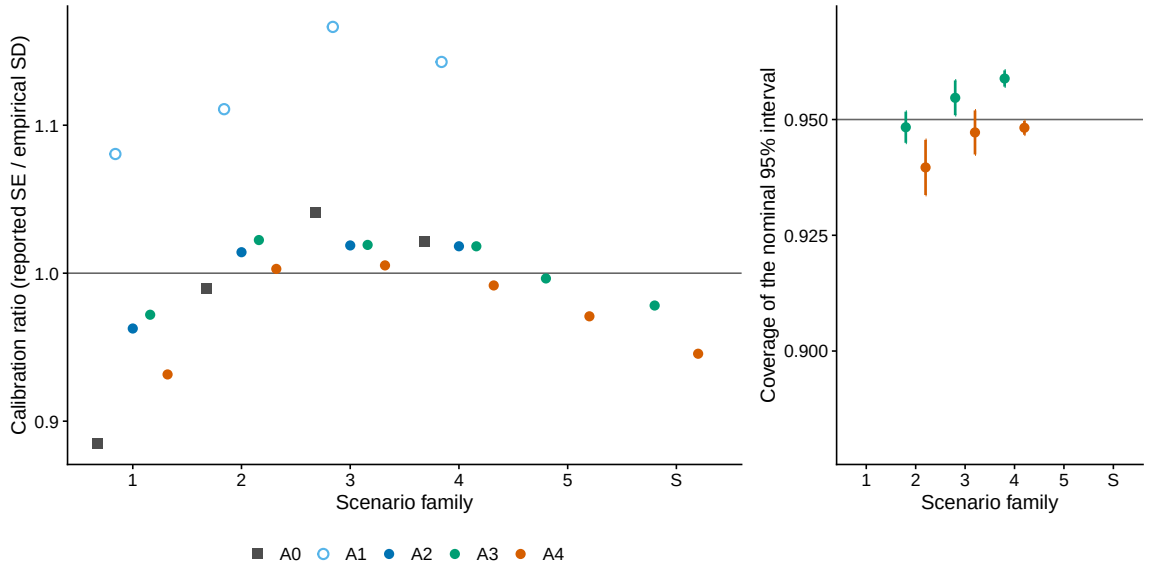


Figure 6: Calibration and coverage. Left: the calibration ratio, the mean reported standard error divided by the empirical standard deviation of the estimates, averaged without weights over the cells of each scenario family, by analysis. A value of one is calibrated and a value above one is conservative. Right: the coverage of the nominal 95 percent interval of A3 and A4 in families 2 to 4, as the mean of the cell lower bounds and upper bounds. The bars span the two bounds. The coverage bounds are not Monte Carlo intervals. Families 1, 5 and S have no coverage output.

Table 7: Coverage of the nominal 95 percent CR2 interval for A3 and A4 over the predictive-only alternative cells of the supplementary study without heavy missingness, on the calibrated scale ($\theta = -c_{bm}\sigma_{BR}$). The Satterthwaite degrees of freedom are not stored per replicate and were recovered from the stored t and p where $|t| \geq 0.5$; the remaining replicates are classified as covering, not covering, or ambiguous, which gives the lower and upper bounds. A3 from '34-referee-reanalysis.R', A4 from '43-g10-coverage.R'.

Method	N	Cells	Mean coverage (lower, upper)	Minimum (lower)
A3	35	15	0.951, 0.958	0.938
A3	70	15	0.951, 0.954	0.915
A4	35	15	0.946, 0.955	0.914
A4	70	15	0.934, 0.941	0.869

Table 8: Scoreboard of the analyses over the scenario families. Families in band counts the families whose pooled null rejection rate lies in 0.04 to 0.06, out of the families in which the analysis was fitted; it is descriptive and does not separate the analyses (Table 9 compares size with the oracle directly). Smallest and largest size are cell-level null rejection rates. Power is the mean over families of the pooled power as a percent of A4 in the same family. The worst power loss is against A4 in points of power, by family and by cell; a negative value is a gain. Coverage is the mean of the nominal 95 percent interval over families with coverage output (A3 and A4 only). A dash means not available; the power columns of A0 are dashed because its test is liberal. A4 is the reference, so its losses are zero.

Analysis	Families in band	Smallest size	Largest size (cell)	Power, percent of A4	Worst family loss, points (family)	Worst cell loss, points (cell)	Coverage
A0	3 of 4	0.034	0.150 (CO, $N = 35$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	–	–	–	–
A1	0 of 4	0.016	0.048 (OL+BDC, $N = 35$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	92.7	6.5 (1)	12.1 ($\rho = 0.9$, $N = 35$, $c_{bm} = 0.30$, $t_{1/2} = 0.5$)	–
A2	2 of 4	0.030	0.074 (OL+BDC, $N = 35$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	96.9	4.1 (1)	9.3 (OL+BDC, $N = 35$, $c_{bm} = 0.45$, $t_{1/2} = 1$)	–
A3	5 of 6	0.032	0.054 (Hybrid, $N = 70$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	100.9	0.3 (2)	2.0 (OL+BDC, $N = 35$, $c_{bm} = 0.30$, $t_{1/2} = 1$)	0.954
A4	6 of 6	0.026	0.056 (Hybrid, $N = 70$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	100.0	0	0.0 ($t_{1/2} = 0.2$ wk, $N = 35$, $c_{bm} = 0.25$, $t_{1/2} = 0.2$)	0.945

3.5 Which analysis to use

Table 8 is a scoreboard over all scenario families. For each analysis it gives the number of families whose pooled null rejection rate lies in the 0.04 to 0.06 band, the smallest and largest cell-level size, the power relative to A4, the worst power loss against A4 by family and by cell, and the mean coverage.

The band counts of Table 8 do not separate the analyses. A4 lies in the 0.04 to 0.06 band in every family. A3, the oracle, misses it in one family, with a pooled rate of 0.0396 in family 4, where A4 has 0.0408. A count that ranks the oracle below A4 on so small a difference measures noise. A paired comparison with the oracle is sharper (Table 9). The difference of A4 from A3 is -0.0008 in family S, the run with independent seeds, with a 90 percent interval

Table 9: Size of A4 against the oracle A3, by family: the paired difference in null rejection rate on the same trials, its paired standard error, and the 90 percent interval. An interval inside ± 0.01 is equivalence by two one-sided tests at the 5 percent level. The paired SE is the replicate-weighted mean of the cell SEs, an upper bound, so the interval is conservative. The margin was chosen after the results were seen. Family S has independent seeds per cell; the cells of families 2 to 4 share random-number streams.

Family	A4 − A3	Paired SE	90% interval	Within ± 0.01
S (independent seeds)	-0.0008	0.0016	-0.0033 to +0.0018	yes
1	+0.0023	0.0047	-0.0054 to +0.0101	no
2	+0.0000	0.0043	-0.0071 to +0.0071	yes
3	-0.0015	0.0041	-0.0083 to +0.0053	yes
4	+0.0012	0.0046	-0.0064 to +0.0088	yes
5 (one cell)	+0.0000	0.0028	-0.0047 to +0.0047	yes

of -0.0033 to +0.0018. That is equivalence within ±0.01. The interval lies inside the margin in 5 of the 6 families. In family 1 it reaches +0.0101, just outside, with a conservative SE.

The largest cell size of A4 is 0.056. A0 lies in the band in 3 of 4, with a largest cell size of 0.150. A1 lies in the band in 0 of 4 and loses up to 12.1 points of power against A4 in a single cell.

For a Hybrid trial with a predictive-only biomarker, A4 is the analysis to use. It needs no half-life, and its size is equivalent to the oracle's within ±0.01 in the run with independent seeds and in 4 of the other five families. The other analyses reach 92.7 to 100.9 percent of its power on average; the analyses above 100 percent, A0 and A3, are respectively liberal and unavailable to an analyst. The conditions and limits are in Section 4.

3.6 Sample size for A4

Table 10 gives the total N at which A4 has 80 percent power, and Figure 7 shows the power curves behind it for the parameter set of this paper. Halving the effect roughly quadruples N. Under this paper's parameter set a Hybrid trial with a 1-week half-life needs about 73 patients at $c_{bm} = 0.4$, 130 at 0.3 and 292 at 0.2. Carryover raises the requirement in the two designs with a discontinuation phase: at $c_{bm} = 0.3$ the Hybrid design needs 67 patients without carryover and 130 at a 1-week half-life, and OL+BDC 88 and 211. The crossover design is almost insensitive to the half-life but needs about twice the Hybrid sample without carryover. Under the starting values the required N is 1.9 to 2.1 times larger, because the response components vary more; the ordering of designs and half-lives is the same.

The binary indicator with CR2 (A2) needs the same N as A4 within a few percent, except at a 1-week half-life in the two designs with a discontinuation phase. There it needs 15 to 18 percent more patients than A4 in the Hybrid design and 22 percent more in OL+BDC, over both parameter sets. That is the gain from modeling carryover expressed in patients. Table S14 gives the N for A2 in every cell.

A4 was also simulated in the Hybrid design under the starting values with the simulation code of the companion paper (2), with its own seeds and grid; the companion paper does not report these runs. A likelihood-ratio test of a difference between the two studies' power curves gives $p = 0.10$ without carryover and $p = 0.91$ at a 1-week half-life, and the two studies' N for 80 percent power differ by at most 6 patients over c_{bm} from 0.25 to 0.40. In runs made with the companion paper's simulation code under the starting values, A4 had power 0.844 to 0.936 at the sample size of that paper's closed-form paired-difference rule, in every cell where that sample size was at most 400 (Table S15). The rule is therefore conservative for A4 under the starting values. Serial correlation matters as much as the half-life: in the same code's runs the Hybrid N at $c_{bm} = 0.25$ is 0.35 to 0.40 times its $\rho = 0.7$ value at $\rho = 0.9$ and 1.13 to 1.40 times at $\rho = 0.5$, over half-lives of 0, 0.2 and 1 week (Table S13).

The curve model is an approximation. Fitting a separate curve for each c_{bm} changes N by a factor of 0.95 to 1.06, and 6 of the 36 curves fail a goodness-of-fit test at the 5 percent level. The intervals of Table 10 cover Monte Carlo error under the model only, so the values are

Table 10: Total N for 80 percent power of A4 at $\alpha = 0.05$, with a 95 percent Monte Carlo interval, by design, carryover half-life (weeks) and c_{bm}, under the parameter set of this paper and under the starting values of the companion paper. From a probit curve in $c_{bm}\sqrt{N}$ fitted to five values of N per c_{bm}, 500 replicates each; the interval does not include the approximation of the curve model (about 6 percent). $\rho = 0.7$, exponential decay, complete data, predictive-only biomarker.

		This paper, c_{bm}			Starting values, c_{bm}		
Design	$t_{1/2}$	0.2	0.3	0.4	0.2	0.3	0.4
CO	0.0	302 (295–310)	135 (131–138)	76 (74–78)	566 (551–582)	252 (245–259)	142 (138–146)
	0.5	302 (295–311)	135 (131–138)	76 (74–78)	571 (556–587)	254 (247–261)	143 (139–147)
	1.0	316 (308–324)	141 (137–144)	79 (77–81)	614 (598–631)	273 (266–281)	154 (150–158)
OL+BDC	0.0	198 (193–203)	88 (86–91)	50 (49–51)	418 (407–431)	186 (181–192)	105 (102–108)
	0.5	292 (285–300)	130 (127–134)	73 (72–75)	612 (595–630)	272 (265–280)	153 (149–158)
	1.0	474 (459–490)	211 (204–218)	119 (115–123)	1010 (975–1048)	449 (434–466)	253 (244–262)
Hybrid	0.0	151 (146–156)	67 (65–70)	38 (37–39)	310 (302–320)	138 (134–142)	78 (76–80)
	0.5	203 (198–209)	91 (88–93)	51 (50–53)	420 (408–432)	187 (182–192)	105 (102–108)
	1.0	292 (284–299)	130 (127–133)	73 (71–75)	613 (597–631)	273 (266–281)	154 (150–158)

best read as accurate to about 6 percent. They assume complete data, the exponential decay form, $\rho = 0.7$ and a predictive-only biomarker.

4 Discussion

4.1 What each rung bought

The ladder was not planned in advance. Each rung answered the question the previous one raised. The recommendation rests on the chain of rungs and on the stress scenarios rather than on any single comparison. Everything below is conditional on the correlation encoding of the companion study, on one parameter set chosen for the simulation core (Table 2), and on the Hybrid design. The analysis model is close to the conditional mean of that Gaussian DGP. Part of what the rungs show is therefore that a correctly specified regressor outperforms a misspecified one. The findings concern which choices cost power or size under that DGP, not the DGP itself.

The direction of rung 3 depends on the encoding. The factorial grid also ran the mean-moderation encoding, in which the biomarker shifts the drug-response mean at on-drug visits only, so no moderation persists after discontinuation. There the binary indicator is the correctly specified regressor. At a one-week half-life exposure weighting changed power by -6.6 points under Hybrid and -5.0 under OL+BDC, against +6.0 and +5.5 under the correlation encoding. Rung 3 therefore pays only when the moderation fades with the drug effect. Which encoding holds for a given drug is an empirical question that these trials have not yet answered.

Rung 1 showed that the starting analysis is liberal and that A1 is conservative. A0 is liberal by a factor of two to three under the crossover and OL+BDC designs and marginally liberal under Hybrid. It differs from A1 in three respects at once. The isolation analyses of Section 3.2.1 separate them. The missing residual correlation accounts for the whole inflation. The

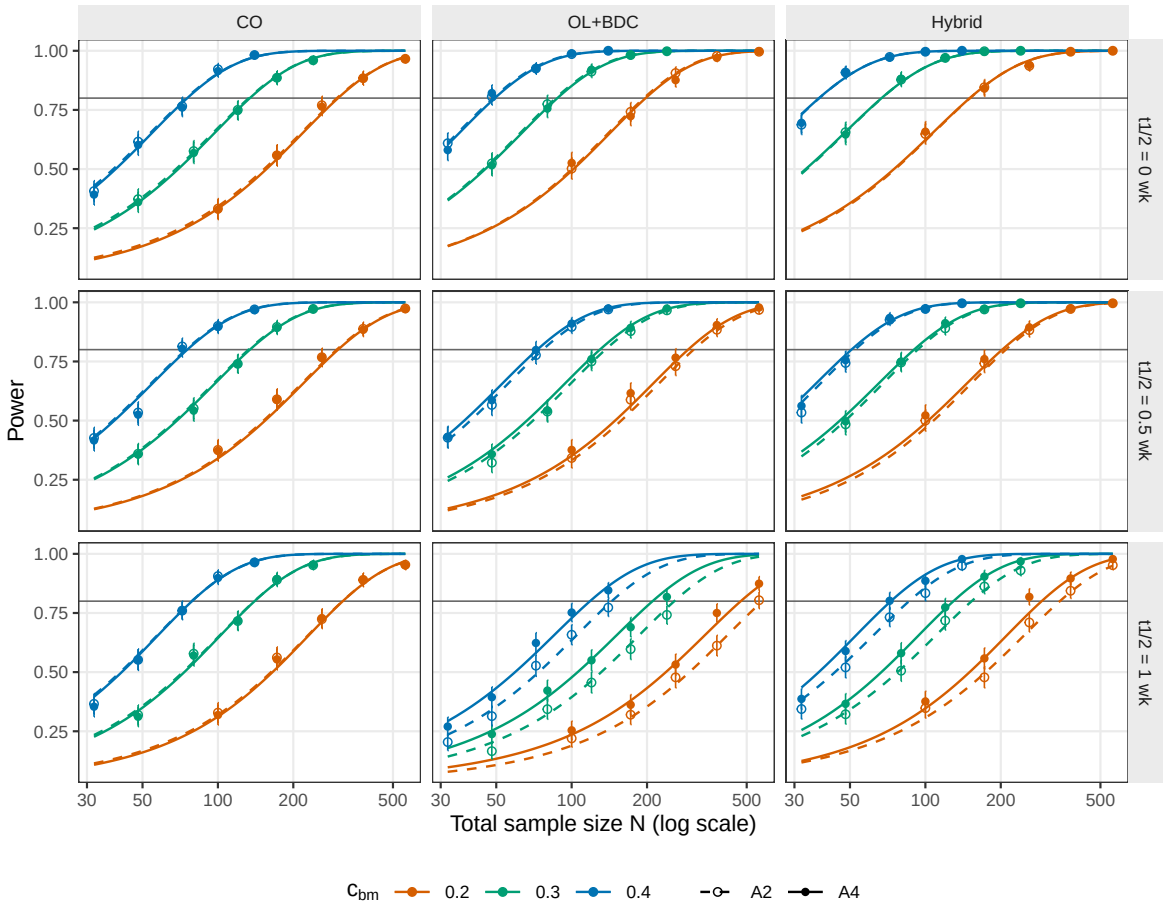


Figure 7: Power of A4 (filled points, solid lines) and A2 (open points, dashed lines) against total N under the parameter set of this paper, by design (columns) and carryover half-life (rows), for $c_{bm} \in \{0.2, 0.3, 0.4\}$. Points are simulated rejection rates (500 replicates) with $2 \times$ the Monte Carlo standard error; lines are the fitted probit curves in $c_{bm}\sqrt{N}$. The horizontal line is 80 percent power. The same figure under the starting values is Figure S1.

software, the degrees-of-freedom method, the expectancy term and the exclusion of patients change size by at most 0.005. A1 costs power against A0 in the base grid, but the power of A0 is the power of a test that rejects too often.

Rung 2 showed that the working-covariance test is conservative by about a quarter of its nominal size. The conservatism comes from a working covariance that does not match the data. A CR2 cluster-robust standard error with Satterthwaite degrees of freedom brings every analysis to within about half a point of nominal size. It changes power by a mean of +2.4 points under Hybrid, +3.3 under OL+BDC and -1.0 under the crossover design.

Rung 3 showed that the choice of exposure representation is a question of power, not validity. Only the representation that changes the regressor carrying the interaction recovers any power. Even at the true half-life, exposure weighting recovered only 15% to 44% of the loss at a one-week half-life under Hybrid and 19% to 25% under OL+BDC. Most of the loss is not recoverable by the analysis and has to be met by the design. The exposure-weighted predictor was within Monte Carlo noise of the binary indicator where it did not gain. The earlier finding that it failed under a crossover design was a coding defect, and we withdraw it. Under this DGP the exposure-weighted predictor at the true decay is the correctly specified regressor. Rung 3 therefore measures the cost of misspecifying the regressor, not a general superiority.

Rung 4 showed that the true half-life, which rung 3 gave the analyst, is not needed. Choosing it by AIC on the model without the interaction (A4) changes power against the oracle A3 by -0.013 to +0.003 and size by -0.002 to +0.002 across the six families. This includes the crossover design, once the selection compares ML fits and the predictor is coded correctly before first exposure. Rung 4 does not show that the selection recovers the half-life. The selected value is spread over the candidates (Table S4). The small power cost means the exposure weight is insensitive to the half-life over this range.

Selecting the half-life with the interaction in the selection model instead raised size above the oracle's in every family (Section 2.4.7). That is why A4 leaves the term under test out of the selection. The scenario grids confirmed the ordering across sample size and effect size. Dropout of 10 to 30 percent of patients, completely at random or dependent on baseline severity, changed neither the size of the CR2 analyses nor their ordering. Only the heavier per-occasion convention, which removes up to 73 percent of the observations, collapsed power. A patient-specific linear trend of 0.3 outcome units per week, unrelated to the biomarker, left size and power unchanged.

Size was evaluated at $N = 35$ and $N = 70$. The working-covariance tests were conservative at both, which points to a working covariance that does not fit rather than to a small-sample effect. The small-sample refinements of CR2 would matter less in larger trials, but the need for a robust standard error would not.

4.2 The recommended analysis for a Hybrid trial

For a Hybrid aggregated N-of-1 trial with a biomarker that is predictive only, simulated under the correlation encoding at $N \leq 70$, we recommend A4:

$$\mathrm{Sx}_{it} \sim bm_i + t + D_{bc,it}(\hat{t}^a_{1/2}) + bm_i{:}D_{bc,it}(\hat{t}^a_{1/2}),$$

with a patient random intercept and `corCAR1` residual correlation. $D_{bc,it}$ equals one on drug, $2^{-t_{sd}/\hat{t}^a_{1/2}}$ off drug after first exposure, and zero before first exposure. $\hat{t}^a_{1/2}$ is chosen by AIC over $\{0.25, 0.5, 1, 2\}$ weeks from ML fits of the model without the interaction, $\mathrm{Sx}_{it} \sim bm_i + t + D_{bc,it}$. The model with the interaction is then fitted by REML at that half-life. The interaction is tested with a CR2 cluster-robust standard error and Satterthwaite degrees of freedom. Two properties recommend it. It requires no pharmacokinetic input from the analyst. Its size is within one paired standard error of the oracle at the true half-life in every family, because the half-life is chosen without reference to the term under test. Two qualifications apply. Its standard error under the alternative is still too small, by a few percent in pooled families and by up to about 13 percent in single cells, and its interval covers about one point below nominal at $N = 70$ (Section 3.4). With a predictive-only biomarker A2 to A4 hold nominal size without the design-expectancy terms in every family examined; the terms matter when the biomarker may also be prognostic, which the companion paper (12) examines. A trial whose biomarker may be prognostic should follow that paper, which pairs the selection rule of A4 with the expectancy terms.

The recommendation transfers to OL+BDC on the evidence of rungs 3 and 4 and the scenario cells, where A4 and A3 lead in the same near-tie. Under the crossover design the analyses are within a few points of one another. A4 has the lowest power of A1 to A4 there, 0.615 against 0.631 for A1 on the base grid, and the binary indicator is as good a choice. The recommendation is stated for the correlation encoding.

4.3 Limitations

The numbers come from a single parameter set chosen for the simulation core (Table 2). It is not calibrated to data, and it differs from the starting values that the companion paper uses. Its placebo-response SD is a fifth of the starting value, so the placebo pathway is weak. Reruns of the main cells under the starting values exist but are not reported here. That set has one trajectory shape per component, one response SD vector, $\rho = 0.7$ except in the ρ grid, and $c_1 = 0.2$. The recommendation comes from one design. Size was evaluated at $N = 35$ and $N = 70$; behavior at larger samples was not examined for a predictive-only biomarker. Absolute power values are therefore illustrative. The ranking, conditional on design, is the more portable result. The correlation encoding is itself an unvalidated modeling assumption of this project. The analysis model is close to its conditional mean. The study therefore cannot show how the ranking would change under a DGP with a different mean structure. Every biomarker simulated here is predictive only; a biomarker that is also prognostic is examined in the companion paper (12). Selecting the half-life costs a few percent of standard-error calibration under the alternative. Coverage of the CR2 interval has been quantified only over the cells of the supplementary study (Section 3.4). A selection-aware interval has not been evaluated. Dropout was examined at 10 to 30 percent of patients, monotone, with the MAR mechanism tied to baseline severity. Baseline severity is not among the analysis variables,

so the mechanism is MAR given the analyzed data only insofar as early outcomes proxy for it. Informative dropout related to the outcome or to treatment response was not simulated. The dropout block has no alternative cell without dropout under the same code. Under the heavier per-occasion convention power collapsed for every specification. The recommendation is therefore for trials that retain most of their observations. The scenario grids use one alternative effect size per scenario. The patient-specific slope variance was simulated at one value, in one Hybrid cell, and with A3 and A4 only; A4 was also fitted with a random slope on time. Cells within a job share random-number streams. Counts of cells inside a band are therefore not independent, and the cross-job comparisons are unpaired. The calibration ratio κ uses the mean rather than the root mean square of the standard errors, which lowers it by 0.01 to 0.03 (Section 2.4.4). The effect-size curve of earlier drafts was not rerun under the corrected code. The sensitivity checks vary one factor at a time, so they can miss interactions between factors.

4.4 Related approaches not evaluated

Three N-of-1 analysis approaches sit outside the family examined. Each would require substantial independent development to enter the comparison. Tang and Landes (20) propose formula-based t-tests that correct a single-subject paired-difference test for an estimated serial correlation. They have no biomarker-moderation term. Schmid and Yang (21) describe a Bayesian hierarchical alternative with partial pooling of individual-level treatment, trend and autocorrelation parameters, and discuss extensions for carryover. Daza (22) frames N-of-1 inference as a counterfactual time-varying-treatment problem. In that framing carryover is confounding to be controlled rather than a predictor to be modeled. Chen and Chen (3) found that a paired t-test matched or exceeded mixed models in simulated N-of-1 series with and without carryover. Araujo, Julious and Senn (23) describe the variance components that a set of N-of-1 trials identifies. These two papers motivated a paired-difference specification in earlier drafts. On the corrected grid it trails the mixed-model analyses under every design and is not pursued. The closest prior work on an exponentially decaying exposure term is the carryover-adjusted parametric model of Gärtner et al. for aggregated N-of-1 trials, with exponential wash-in and wash-out (6). Its estimand is the treatment effect, not the biomarker interaction. The biomarker interaction is where this study's contribution lies. Blackston et al. (5) show inflated Type I error for aggregated N-of-1 analyses of the treatment effect that ignore moderate or strong carryover. The interaction behaves differently. A1 ignores carryover and is conservative, and the inflation of A0 comes from its missing residual correlation (Section 2.4.2). For the interaction, ignoring carryover costs power rather than validity. Percha et al. (4) give design recommendations from simulated single-series N-of-1 trials with carryover and wash-in. Zucker et al. (24) set out the methodology for combining N-of-1 trials into population estimates. The case against a single lagged term as a carryover adjustment in crossover trials is classical (25–27).

5 Conclusions

1. **Rung 0 to 1: do not use the starting analysis as is.** It is liberal by two to three times nominal under the crossover and OL+BDC designs and marginally liberal under Hybrid. It differs from A1 in its residual correlation structure, an expectancy term and its software. A1 is conservative and costs power against A0 in the base grid.
2. **Rung 2: use a CR2 cluster-robust standard error.** The working-covariance test rejects at about 0.037 under the null. CR2 with Satterthwaite degrees of freedom brings every analysis to within about half a point of nominal size. It changes power by a mean of +2.4 points under Hybrid and +3.3 under OL+BDC, and by -1.0 under the crossover design.
3. **Rung 3: represent residual exposure in the regressor.** An exposure-weighted predictor at the true half-life recovers part of the power that carryover removes under the Hybrid and OL+BDC designs, 15% to 44% and 19% to 25% at a one-week half-life. The rest has to be met by the design, for example with more participants. It is within Monte Carlo noise of the binary indicator where it does not gain.
4. **Rung 4: let the data choose the half-life, on the model without the interaction.** AIC selection over four candidates, comparing ML fits of the model without the interaction and refitting the interaction model at the selected half-life by REML, removes the need for the analyst to supply a half-life. Paired against the oracle at the true half-life, it changes size by -0.002 to +0.002 and power by -0.013 to +0.003 across the six families. The selected value is spread over the candidates, so the small power cost reflects the insensitivity of the predictor to the half-life rather than recovery of it. Leaving the interaction out of the selection keeps the size of the oracle; selecting with it in the model raised size in every family.
5. **Recommendation.** For a Hybrid aggregated N-of-1 trial with a biomarker known to be predictive only, simulated under the correlation encoding at $N \leq 70$, use A4: the exposure-weighted predictor at the half-life chosen by AIC on the model without the interaction, with CR2 inference. Its size is equivalent to the oracle's within ± 0.01 in the run with independent seeds and in 4 of the other five families. Against A2, the binary indicator with CR2, A4 gains 1.3 points of power on average in the Hybrid cells, up to 7.1 at long half-lives, and loses at most 1.0 point in the 6 cells where it is behind. Among the analyses fitted in every family that do not need the true half-life, A4 is the most powerful one whose size is within Monte Carlo error of the oracle's. Under dropout of 10 to 30 percent of patients, completely at random or dependent on baseline severity, A4 keeps its size and, in all but one cell, its power advantage over the binary indicator. It should be accompanied by an effort to retain patients, because heavier missingness costs every analysis power.
6. **A biomarker of unknown role.** Whether a candidate biomarker is also prognostic is rarely known before a trial and is hard to test within it. A4 omits the design-expectancy terms, which protect the test when the biomarker is prognostic through a pathway that changes over time. The default for a biomarker of unknown role is therefore the analysis of the companion paper (12). This paper supports the parts of that analysis it shares

with A4: the exposure-weighted predictor, the choice of half-life on the model without the interaction, and the CR2 test.

7. **Sample size.** Halving the effect roughly quadruples the required N. Under this paper's parameter set a Hybrid trial with a 1-week half-life analyzed with A4 needs about 73, 130 and 292 patients at $c_{bm} = 0.4$, 0.3 and 0.2, and the binary indicator about 15 to 18 percent more (Table 10). The requirement is roughly twice as large under the starting values and depends strongly on the serial correlation.
8. **Limits.** The recommendation is conditional on the correlation encoding, on one parameter set (Table 2), on one design and on a biomarker that is predictive only. Size was evaluated at $N = 35$ and $N = 70$. The analysis model is close to the conditional mean of the DGP, so part of what the rungs show is that a correctly specified regressor outperforms a misspecified one.

Glossary

- **Analysis codes (A0 to A4).** The five analyses of Table 3, numbered by rung. The companion papers and earlier drafts call them E0, G1, G6, G8 and G10. Stored data files label them `E0`, `E1`, `E1cr2`, `E2cr2` and `E7null_cr2`.
- **A4.** The recommended analysis: the exposure-weighted predictor at the analysis half-life chosen by AIC on the model without the interaction, refitted by REML with the interaction and tested with CR2 (G10 in the companion papers).
- **Oracle benchmark (A3).** An analysis that uses the true data-generating half-life. It is not available to an analyst and is not a candidate analysis. It shows what the selection step costs.
- **Slope variance.** A data-generating option added for the referee-variant runs (`sd_slope`). It adds a patient-specific linear trend, unrelated to the biomarker, with the stated standard deviation per week.
- **Isolation analyses (E0n, E0nc, E1nc).** Three analyses that change one difference between A0 and A1 at a time. E0n refits the formula of A0 with `nlme` and no residual correlation. E0nc adds `corCAR1` to E0n. E1nc uses the formula of A1 with `nlme` and no residual correlation. A1 is E1nc with `corCAR1`.
- **Ladder.** The ordered set of analyses of Table 3, from the starting analysis (rung 0) to the analysis with an AIC-selected half-life and CR2 inference (rung 4).
- **Rung.** One analysis of the ladder. Each rung changes one thing from the rung before, except the step from rung 0 to rung 1.
- **Scenario family.** A set of simulation cells that share one departure from the base grid. The families are numbered 1 to 5, with family S for the Hybrid run with independent seeds (Table 1).
- **Biomarker-response correlation (c_{bm}).** The on-drug correlation between the biomarker and the drug-response component. It is the interaction estimand of this study. Stored data and code refer to it as `c_bm` or `coupling`.
- **Carryover half-life ($t_{1/2}$).** The time after discontinuation at which the residual drug effect, and under the correlation encoding the biomarker-response correlation, has fallen

to half its on-drug value. The analysis half-life $t^a_{1/2}$ is the value the analyst assumes or selects.

- **Common random numbers.** Fitting every specification to the same simulated dataset within a cell, so that differences between specifications are paired and have reduced Monte Carlo variance.
- **CO (crossover).** The two-period blinded crossover design of Section 2.2.
- **Cluster-robust standard error (CR2).** The bias-reduced cluster-robust sandwich standard error of Bell and McCaffrey, which estimates the variance of the interaction coefficient from between-patient variation. It remains valid when the working covariance is misspecified. It is paired with Satterthwaite degrees of freedom.
- **Working-covariance standard error.** The standard error that `nlme::lme` reports, computed under the working covariance: a patient random intercept with `corCAR1` residual correlation. It is correct only if that covariance is the true one. A test that uses it is called a working-covariance test. Each cluster-robust analysis fits the same model and changes only the standard error, the reference distribution and the p-value.
- **Correlation encoding of the interaction.** The data-generating convention in which the biomarker-treatment interaction is a correlation between the biomarker and the drug response that is present on drug and absent or decayed off drug. Code and data files label it `mvn` or covariance moderation.
- D_{bc}**.** The exposure-weighted predictor: one on drug, $2^{-t_{sd}/t^a_{1/2}}$ off drug after first exposure, zero before first exposure.
- **Expectancy (**D_e**,** η_t**).** The participant's probability of being on drug by design: one during open-label periods and one half during blinded periods. It scales the placebo-response component.
- **Hybrid design.** Open-label titration, blinded discontinuation, then a brief crossover (Section 2.2).
- κ **(calibration ratio).** The mean reported standard error (working-covariance or cluster-robust, whichever the analysis uses) divided by the empirical standard deviation of the estimates across replicates. A value of one is correctly calibrated.
- **MCSE.** Monte Carlo standard error, the standard error of a simulation estimate that arises from the finite number of replicates.
- **Natural-history component (TV).** The latent component of the outcome that follows the untreated time course.
- **OL+BDC.** Open-label titration followed by blinded discontinuation (Section 2.2).
- **Placebo-response component (PB).** The latent component of the outcome that responds to expectancy rather than to the drug.
- **Predictive-only biomarker.** A biomarker that changes the drug response and nothing else: it is uncorrelated with baseline severity, the natural-history component and the placebo-response component. Every cell of this study has such a biomarker.
- **Recommended DGP** (`refined` in code and data labels). The data-generating process of the companion study, in which the biomarker-response correlation decays after discontinuation, the within-component correlation is separable AR(1), carryover decays from the last on-drug mean, and a non-positive-definite covariance matrix stops the

simulation.

- **Satterthwaite degrees of freedom.** An approximate degrees of freedom for the t reference distribution of a cluster-robust test, computed from the variance of the variance estimate.
- **Separable AR(1) structure.** The same first-order autoregressive correlation for every component across occasions, with a constant between-component correlation at every lag.
- **Starting analysis (A0).** The analysis published with the starting DGP: `lme4::lmer` with a random intercept, no residual correlation structure, a binary on-drug indicator, and Satterthwaite p-values. Paper 01 calls it the binary-indicator analysis.
- **Starting DGP.** The data-generating process of Hendrickson et al. (2020), the starting point of the companion study.
- **Supplementary study.** A later simulation study that added the shape, half-life, serial-correlation and dropout cells of families 2 to 4. Code and data files label it `gap-fill`.
- **Two-MCSE band.** The interval of plus or minus two Monte Carlo standard errors around the nominal 0.05 rejection rate, which is $[0.031, 0.069]$ at 500 replicates, $[0.036, 0.064]$ at 1,000 and $[0.040, 0.060]$ at 2,000.

6 Declarations

Ethics approval and consent to participate. Not applicable. This is a simulation study with no human participants or patient data.

Consent for publication. Not applicable.

Availability of data and materials. The data and code supporting the findings of this study are available in the pmsimstats-ng project repository (`https://github.com/rgt47/pmsimstats-ng`). A tagged release with a persistent identifier will be deposited at submission. The commit reported here should be cited until then. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`. Simulation drivers are at `analysis/scripts/carryover-sensitivity/`. Exact paths are listed in the Reproducibility section below. The stored summaries also contain cells from a mean-shift encoding of the interaction (`meanmod` and `B3-meanmod` labels) and a linear decay form. Both are outside the main scope of this manuscript; the mean-shift cells of the factorial grid enter only the Discussion's paragraph on the direction of rung 3. The data dictionary in the repository marks them as such.

Competing interests. To be completed by the authors. Any author of this paper who is also an author of the published framework (1) must be disclosed here.

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Reporting. This is a simulation study and reports no trial. The CENT 2015 extension of CONSORT (28) governs the reporting of any N-of-1 trial that applies the recommended analysis.

7 Reproducibility

This manuscript was rendered inside the project's zzcollab Docker container, with the image hash in `Dockerfile` and the package set in `renv.lock`. The simulations ran under R 4.6.0 on the project's simulation host. The study uses `nlme` for the continuous-time linear-mixed analysis, `clubSandwich` for CR2 inference, `lme4` and `lmerTest` for A0, and `parallel` for the parameter-grid sweep. No simulation code runs during knitting. Knitting loads pre-computed summaries only.

Drivers and data, by table and figure. All paths are under `analysis/scripts/carryover-sensitivity/` (scripts) and `analysis/data/` (summaries).

- Factorial grid of job 26 (decay-form comparison and relative losses of A1 in Section 3.2.3): `01-run-factorial.R` (540 cells, 500 replicates, master seed 20260415; rerun 2026-10-25 under the corrected code) and `02-summarise-grid.R` producing `02-grid-summary.rds`.
- Corrected 30-cell grid of job 27: `27-run-tier1-g-vs-f-e0.R` (30 cells, 500 replicates, 2026-10-25) producing `carryover-sensitivity/tier1-g-vs-f-e0/{summary,all}.rds`. Table 5 and Figure 5 are computed from the per-replicate `all.rds` by the chunk of Section 3.2.3 and by `40-stage-power-figure.R`, which writes `analysis/figures/02-power-by-half-life.` Table S1 and Figures 3 and 4 (via `45-ladder-heatmaps.R`) are drawn from the summary. A4 in all of them comes from the backfill below. The size diagnostic of Table S3 is block A of the supplementary study below.
- Scenario cells: `06-g-vs-f-comparison.R` (9 of its 15 cells are reported; 500 replicates, 2026-10-23) producing `carryover-sensitivity/g-vs-f-comparison/summary.rds`.
- Supplementary study (labeled `gap-fill` in code and data): `32-run-g9-gap-fill.R` under the working protocol `32-gap-fill-protocol.md`, run as four simrun jobs (blocks A, B and C; jobs 31, 32 and 33 on the primary host and job 40 on a second, 2026-10-26 to 2026-10-27, at commit `ca2eef2` or a code-identical successor, except job 33, which reran the dropout cells at commit `fc7ee08` under the patient-level convention), assembled by `33-assemble-gap-fill.R` into `carryover-sensitivity/g9-gap-fill/summary.rds` with each job's own outputs under `g9-gap-fill/job-NN/`. Tables S3, S9, S10 and 6 are drawn from it. The per-occasion dropout cells of job 31 and the prognostic cells of the companion paper (12) are in the same file. The pre-flight eigenvalue of every cell is in `preflight.rds`.
- Corrections: the paired check of the pre-exposure coding defect is `26-co-dbc-fix-paired.R` (`carryover-sensitivity/26-co-dbc-fix-paired-reps500.rds`); the cross-factor constant correction and the stop-on-non-PD default are in `simulation-core.R` and `implementations/tidyverse/R/functions.R`.
- Reanalysis of the per-replicate output for the 2026-10-27 referee report (calibration and coverage at the alternatives, the heavy-missingness dispersion):

`34-referee-reanalysis.R` producing the CSV files in `carryover-sensitivity/referee-reanalysis/`, from which Tables 7 and S11 are drawn. Table S4 is computed from the backfill below. The A4 rows of Table 7 come from `43-g10-coverage.R`, which applies the same method to the A4 refits of `41-run-g10-backfill.R`, checks that its bounds for the selection variant of Section 2.4.7 reproduce `m3-coverage.csv` exactly, and writes `m5-coverage-g10.csv`.

- Referee-variant runs: `36-run-referee-variants.R` (checked by `35-check-new-variants.R`) adds, with an independent seed per cell derived from the cell label, A4 (stored as `E7null_cr2`) beside A3 and the selection variant of Section 2.4.7 (stored as `E7cr2`) in the Hybrid null and alternative cells (blocks N and A) and a patient-specific slope variance (`sd_slope`, block S). Run as simrun jobs 34, 35 and 37 on the zeus host (8 cores) at commit `086f826` on 2026-10-27 (3.5 hours, 42 minutes and 8 minutes), producing `carryover-sensitivity/referee-variants/block-{N,A,S}/` with `summary.rds`, `selection.rds`, `paired.rds`, `preflight.rds`, `meta.rds` and the job's `provenance.yml`. Tables S5, S6 and S12 are drawn from them. Because these cells have their own seeds, comparisons between them and any other stored cell are unpaired. The runs also fitted the A3 formula at a fixed analysis half-life (`E3fix_cr2`), which is not reported. The null cells with a prognostic placebo pathway (block N) belong to the companion paper and are not reported here.
- A4 in the other families: `41-run-g10-backfill.R` regenerates each stored trial of jobs 27, 21, 32, 33 and 37 from its stored seed and refits the stored selection variant (`E7cr2`) and fits A4 to it (2026-10-31), writing `carryover-sensitivity/g10-backfill/`. The refit of `E7cr2` is the pairing check: a replicate is used only if it reproduces the stored fit, which held in 54,989 of 55,000 replicates (`guard.rds`).
- A4 with a random slope on time: `44-g10-random-slope.R` regenerates the block-S trials from their seeds and fits A4 with a random intercept and with a random slope, recording whether a failure came at the half-life selection or the final fit. Run as simrun job 5 (commit `2c9dadd`, 37 minutes) on the julia host (6 cores) on 2026-11-09, writing `carryover-sensitivity/g10-random-slope/` (`all.rds`, `summary.rds`, `guard.rds`). The random-intercept fit is the pairing check against the stored block-S fit: it reproduced the stored estimate within 10^{-6} in 1,997 of 1,998 replicates. Each exception chose a different half-life. The same script with `--g9` refits the selection variant (job 7, `g9-random-slope/`), which is not reported.
- Ladder tables and figures: `37-ladder-tables-figures.R --scope predictive` reads only stored data, in a few seconds, and writes the CSV files in `carryover-sensitivity/ladder-predicti` and the three figures `analysis/figures/02-ladder-fig{1-size,2-power,3-calibration}-predictive` (Tables 3, 4 and 8, Figures 1, 2 and 6). `37-ladder-README.md` documents its definitions and known limits.
- Sample size for A4: `46-run-sample-size.R` fits A4 and A2 over the N grid of Section 2.6, 500 replicates per cell, as six simrun jobs on 2026-11-11 (one per parameter set and design: athena 21, joe 14, zeus 66 and 67, mercury 53 and 54; commits `5819e1d` and `431f126`, which differ only in paper 01 files), writing `carryover-sensitivity/sample-size-<set>-<design>/`. `47-sample-size-curves.R`

fits the probit curves and writes `sample-size-n80.csv`, `sample-size-fit.csv`, `sample-size-sepcheck.csv` and the figures `02-sample-size-curves-<set>`. `48-reconcile-sample-size-paper01.R` compares the Hybrid run under the starting values with runs made with the companion paper's driver `19-g9-planning-and-fig4b.R` and writes `sample-size-reconcile.csv`. Table S13 reads that driver's `n80-g10.csv` in `published-planning/block-N/`, `published-planning/rho0.5/block-N/` and `published-planning/rho0.9/block-N/`; Table S15 reads its `published-planning/planning-g10.csv`. These runs are not reported in the companion paper.

- Isolation of the first rung: `38-run-e0-isolation.R` fits the chain A0, E0n, E0nc, E1nc and A1 to the same trials on the 30-cell grid at 1,000 replicates (simrun job 42) and writes `analysis/data/carryover-sensitivity/e0-isolation/{all,summary}.rds`. `39-check-e0-isolation.R` checks the fitted models and reproduces A0 and A1 of job 27 on a subset. Section 3.2.1 reads both files, and `37-ladder-tables-figures.R` reads `all.rds`, so the isolation run must precede the ladder script.

Table and figure sources by job. Job 27 supplies Table 5, Figure 5, Table S1 and the two heatmaps. Job 26 supplies the decay-form comparison, the relative loss of A1 and the mean null rejection rate quoted in Section 3.2.3.

The fitting functions for every specification (`fit_spec_s7`, `cr2_extract`, `select_half_life_aic`, `fit_spec_e0`, `simulate_cell_combined`, `simulate_cell_g10`) are in `simulation-core.R`. Specifications are stored under `E`-prefixed codes (`E1`, `E2`, `E3`, `E7`, `E9` and their `cr2` variants, and `E7null_cr2`) in every data file. This manuscript displays them as A0 to A4 at render time through `to_a()` in `spec-labels.R`; the companion papers display them as G codes. Output archived before 2026-07-31 stores the specifications as `A1`, `A2`, `A3`. Those legacy stored codes are unrelated to the A0 to A4 labels of this manuscript. The migration is documented in `analysis/report/NOTATION.md`. The exploratory sensitivity blocks of earlier drafts and their figures are documented in the `archive/` directory of this manuscript, including the earlier supplement (`archive/supplement-2026-10-21.Rmd`). The scripts that produced them are retained under the same directory as above.

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